

Block 3 - Chapter 2: Regulations, Maintenance Forms, Records & Publications

Plain-English Study Guide - read this in place of slogging the whole chapter

AIM Block 3 - first chapter read in class (starts Mon 2026-06-15) | Built for Matthew | Source of truth: **FAA-H-8083-30(B), Chapter 2** (the exact assigned PDF), chapter pages 2-1 to 2-46. | No quiz / no flashcards by request - summary only. Page refs like "p. 2-14" point into the chapter PDF.

How to use this: This chapter is mostly *memorization of names, numbers, and "which document does what."* There is very little hands-on skill here - it is the legal/paperwork backbone of being a mechanic. I define every term the first time it shows up, give a plain analogy, then box the must-memorize items in red. Hit the red boxes hard; those are where exam questions live.

1. The Big Picture: Where the Rules Live (Title 14 CFR)

CFR = Code of Federal Regulations. Think of it as the giant rulebook for the whole U.S. government, split into **50 numbered "Titles."** Each Title is one subject area.

Title 14 is the one we care about: **"Aeronautics and Space."** That is where all the FAA's aircraft rules live. (p. 2-1)

Analogy: Title 14 is like the "Aviation" shelf in a giant library. Within that shelf are individual books called **Parts** (Part 21, Part 43, Part 91, etc.). Within each book are **sections** (like 43.9, 91.417).

How Title 14 is laid out

- Title 14 has **five volumes**. The **first three volumes** (containing **75 active regulations / Parts**) are the **FAA**. Volume 4 = Office of the Secretary of DOT + Commercial Space. Volume 5 = NASA. (p. 2-1)
- The FAA's airworthiness rules sort into **three categories: (1) Administrative, (2) Airworthiness Certification, (3) Airworthiness Operation**. (p. 2-1)
- Old name: **"FARs" = Federal Aviation Regulations**. The FAA now says **"14 CFR part XX"** instead, because **Title 48** ("Federal Acquisitions Regulations") also abbreviates to "FAR" and caused confusion. (p. 2-1)
- **Most Part numbers and sections are ODD-numbered** on purpose - so the FAA can slot in new even-numbered rules later (you'll see this with 43.10, 43.12, 135.422 - "note the even number" = added later). (pp. 2-1, 2-16)
- **SFAR = Special Federal Aviation Regulation:** a narrow rule for one unique situation, usually with a time limit. The SFAR number is just a sequential count - it has no link to the rule it attaches to. (p. 2-1)

MUST KNOW - Section 1

- **14 CFR = Title 14 of the Code of Federal Regulations** = where ALL FAA rules live ("Aeronautics & Space").

- **Three categories of regs:** Administrative, Airworthiness Certification, Airworthiness Operation.
- **The 3 BIG regulations that govern airworthiness: Part 21** (certification of products/parts), **Part 43** (maintenance), **Part 91** (operating rules). Memorize these three together. (p. 2-1, 2-2)
- **SFAR** = Special FAR = narrow, usually time-limited rule.

2. Part 3 - General Requirements (What "Airworthy" Means)

Part 3 is short but huge: it finally **defined the word "airworthy" in a regulation**. Before this, the term was only implied (e.g., in Part 21 section 21.183 and on the airworthiness certificate). (p. 2-1)

Airworthy = the aircraft **(1) conforms to its type design AND (2) is in a condition for safe operation**. Both halves are required. (14 CFR 3.5; p. 2-1)

Two more Part 3 definitions:

- **Product** = an **aircraft, aircraft engine, or aircraft propeller**. (Just those three.)
- **Record** = basically **any way information is preserved** - paper, microfilm, dataplate, stamped mark, bar code, or electronic format.

Two kinds of airworthiness (these two phrases run through the whole chapter):

- **Original airworthiness** = getting the design/aircraft approved and certified the first time (Part 21 land).
- **Recurrent airworthiness** = keeping it airworthy day-to-day over its life (Part 43 + Part 91 land). (pp. 2-1, 2-3)

MUST KNOW - Section 2

- **AIRWORTHY** = **conforms to type design + in condition for safe operation**. Both parts. This is the single most-tested definition in the chapter. (Part 3.5, p. 2-1)
- **Product** = **aircraft, engine, OR propeller** (only those three).
- **Original airworthiness** (Part 21) vs. **recurrent airworthiness** (Parts 43 & 91).

3. Regulation vs. Guidance (the most important distinction in the chapter)

Analogy: A **regulation** is the **LAW** - the speed limit sign. **Guidance** is the driver's-ed manual that tells you a good way to obey the law. You **MUST** obey the law. You don't have to follow the manual word-for-word, but following it keeps you legal and safe.

	REGULATION (the law)	GUIDANCE (advice)
Examples	14 CFR Parts (21, 43, 91...), Airworthiness Directives (AD)	Advisory Circulars (AC), SAIBs, FAA handbooks, manufacturer Service Bulletins
Binding?		

	Mandatory - you can be violated for not following it	Not mandatory - "acceptable" methods, NOT the only methods... <i>unless</i> it's referenced into a reg
Key wording	"No person may...", "shall", "must"	"acceptable to the Administrator", "recommends", "guidance"

The trap: Guidance becomes mandatory if it is "**incorporated by reference**" into a regulation. Example: an **AD** is mandatory (it's a reg, Part 39) even though it's printed in the Federal Register, not in the rulebook itself. (pp. 2-24, 2-25, 2-26)

MUST KNOW - Section 3

- **Regulations = mandatory (the law). Guidance = advisory (not mandatory) UNLESS referenced into a reg.**
- **AC = advisory** (not binding). **AD = mandatory** (it IS a regulation - Part 39).
- The phrase "**acceptable to the Administrator**" signals guidance/acceptable data, NOT a hard requirement.

4. The Three Big Regulations - Up Close

Part 21 - Certification Procedures for Products and Articles

This is the "**birth certificate**" Part - how a design gets approved. It covers **Type Certificates (TC)**, **Supplemental Type Certificates (STC)**, **production certificates**, **airworthiness certificates**, and import/export approvals. Also **PMA** (Parts Manufacturer Approval) and **TSO** (Technical Standard Order) parts. Its biggest job is **original airworthiness**. (pp. 2-3, 2-4)

- **Section 21.50** = "Instructions for Continued Airworthiness." Any **STC made after 1981** must come with a full set of **ICA (Instructions for Continued Airworthiness)** - the maintenance/inspection instructions for the altered part. ICA is built from a **16-subject checklist**. (pp. 2-3, 2-9)

Part 43 - Maintenance, Preventive Maintenance, Rebuilding, and Alteration

This is the heart of your job. It is the standard for maintaining every civil aircraft registered in the U.S. It has **13 rules and 6 appendices**. (p. 2-8) (Full detail in Section 7 below.)

Part 91 - General Operating and Flight Rules

This is an **operating** Part aimed at the owner/operator/pilot, but the mechanic must know it because **Subpart E** (sections 91.401-91.421) is all about maintenance, preventive maintenance, and alterations. Section **91.403(b)** ties Part 91 directly to Part 43: no one may do maintenance except as Part 43 requires. **91.7(a)**: "No person may operate a civil aircraft unless it is in an airworthy condition." (pp. 2-10, 2-21)

Quick map of the other Parts you'll see named (don't memorize all - just recognize them): Part 1 = Definitions; Part 23 = small airplanes ($\leq 12,500$ lb, or commuter $\leq 19,000$ lb / ≤ 19 seats); Part 25 = large/transport airplanes ($> 12,500$ lb); Part 27 = small rotorcraft ($\leq 7,000$ lb); Part 29 = large rotorcraft ($> 7,000$ lb); Part 33 = engines; Part 35 = propellers; Part 39 = ADs; Part 45 = ID/registration marking; Part 47 = aircraft registration; Part 65 = mechanics & other airmen; Part 121/125/135 = air carriers/operators; Part 145 = repair stations; Part 147 = AMT schools; Part 183 = FAA designees. (pp. 2-3 to 2-19)

MUST KNOW - Section 4

- **Part 21** = certification (TC, STC, PMA, TSO) - mostly ORIGINAL airworthiness. **21.50** requires ICA.
- **Part 43** = maintenance - "the heart of aviation maintenance"; **13 rules + 6 appendices**.
- **Part 91** = operating rules; **Subpart E (91.401-91.421)** is the maintenance subpart; **91.403(b)** forces compliance with Part 43; **91.7(a)** = no operating unless airworthy.
- **ICA = Instructions for Continued Airworthiness** (16-subject checklist); required on STCs made after 1981.
- **Weight cutoffs:** Part 23 small airplane = 12,500 lb or less; Part 25 large = over 12,500 lb; Part 27 rotorcraft = 7,000 lb or less; Part 29 = over 7,000 lb.

5. Civil Air Regulations (CAR) - the Old Rulebook

Before today's 14 CFR existed, aircraft were certified under the **CARs (Civil Air Regulations)**, the rules of the 1940s-60s. Old aircraft are still **legally certified under the CAR they were built to**, so a mechanic may still need a CAR as the reference for older airplanes. (pp. 2-23, 2-24)

- **CAR 3** - "Airplane Airworthiness - Normal, Utility, Aerobatic, and Restricted Purpose Categories." It is the **ancestor of today's Part 23**. Examples of CAR 3 aircraft: Piper PA-22/28/32; Cessna 182, 195, 310. (p. 2-23)
- **CAR 4a** - "Airplane Airworthiness" (originated 1936, last amended Dec 15, 1952). It and CAR 4b became the regs for **large aircraft**. So: **CAR 3 = small aircraft, CAR 4a/4b = large aircraft**. (p. 2-23)

Watch the trap: "CAR" has **two meanings** - Civil Air Regulations (old U.S.) AND Canadian Aviation Regulations. Know which one a question means. (p. 2-23) Also: a **CAM (Civil Aeronautics Manual)** contained BOTH regulation and advisory material in one document. (p. 2-2)

MUST KNOW - Section 5

- **CAR 3 -> Part 23 (small aircraft). CAR 4a / 4b -> large aircraft** (CAR 4b is the ancestor of Part 25).
- Old aircraft are still certified to the CAR they were built under - it can still be the maintenance reference.
- **"CAR" is a double meaning:** Civil Air Regulations OR Canadian Aviation Regulations.

6. Suspected Unapproved Parts (SUP)

There are **four kinds of parts** (memorize this list - it's a classic): (p. 2-24)

1. **Good part, good paperwork** -> approved, install it.
2. **Good part, BAD paperwork** -> **SUP** (suspected unapproved).
3. **BAD part, "good" (bogus) paperwork** -> **SUP** (suspected unapproved).

4. **Bad part, bad paperwork** -> unapproved - NEVER install.

The **middle two** are the **Suspected Unapproved Parts (SUP)** - if EITHER the part OR its paperwork is questionable, it's a SUP.

What to do with a SUP: tell the shop foreman / supervisor / quality person; **segregate and quarantine** the part until proper disposition is decided; contacting the manufacturer helps gather facts. Reference: **AC 21-29**, "Detecting and Reporting Suspected Unapproved Parts." Report a SUP at www.faa.gov. (p. 2-24)

MUST KNOW - Section 6

- **4 part types**; the **middle two (good part/bad paperwork & bad part/bogus paperwork) = SUPs.**
- **SUP action: SEGREGATE and QUARANTINE**, notify supervisor/quality, then determine disposition.
- Reference doc = **AC 21-29.**

7. Inside Part 43 - The Maintenance Rules That Run Your Career

Here is what each Part 43 section actually does, in plain English. (pp. 2-14 to 2-20)

Section	What it does (plain English)
43.1	Applicability. Part 43 does NOT apply to an aircraft that has only ever held an experimental certificate (pink Form 8130-7). But if a Standard-cert aircraft is temporarily put into experimental, Part 43 still applies. (p. 2-14)
43.2	Defines " overhauled " vs " rebuilt ." Both require: disassembled, cleaned, inspected, repaired as necessary, reassembled, tested. Difference = how it's tested: OVERHAUL = tested to manufacturer's service limits (used-part limits); REBUILT = tested to same tolerances as a NEW part . (p. 2-14)
43.3	Who MAY PERFORM maintenance - 9 persons (mechanic, repairman, person under supervision, repair station, air carrier, pilot doing preventive maint. on own aircraft, etc.). (pp. 2-14, 2-15)
43.5	Approval for Return to Service (RTS) - requires a proper record entry per 43.9 or 43.11; may need Form 337 for a major repair/alteration. (p. 2-15)
43.7	Who MAY APPROVE for RTS - 7 persons . Note: a repairman may perform work but may NOT approve RTS . (p. 2-16)
43.9	Content/form of maintenance record entries (NOT inspections). The What / When / Who rule. (pp. 2-16, 2-17)
43.10	Disposition of life-limited parts (note even number - added 2002). Defines "life-limited part" & "life status"; 7 control methods. (p. 2-16)
43.11	Content/form of inspection record entries. Also What/When/Who. If item FAILS, a list of discrepancies must be given to the owner . (p. 2-17)
43.12	Falsification of records = grounds for suspension/revocation of certificate. "My signature lasts a lifetime." (p. 2-17)
43.13	Performance Rules - use the methods in the manufacturer's manual / ICA, or other methods acceptable to the Administrator. Tie-in to AC 43.13-1 / -2. (pp. 2-17, 2-18)
43.15	Extra inspection rules - rotorcraft items, annual/100-hour (checklist required), progressive. Defines " routine " (visual, no disassembly) vs " detailed " (thorough, with disassembly). (p. 2-18)
43.16	Airworthiness Limitations - must follow the Airworthiness Limitations section of the manual/ ICA. (p. 2-18)
43.17	Maintenance on U.S. products by certain Canadian persons (AME/AMO) under the U.S.-Canada agreement. (p. 2-18)

Repair vs. Alteration, and Major vs. Minor (the decision that drives your paperwork)

Before any job, ask three questions: **(1)** What am I going to do? **(2)** How does the FAA classify it? **(3)** What documentation does it need? (p. 2-15)

- **Repair** = returns the aircraft to its **previous/unaltered** condition (new tires, brakes, magnetos, even a whole replacement engine that's the correct one). (p. 2-15)
- **Alteration** = **changes/modifies** the aircraft from its previous state (winglets, new avionics, an engine NOT listed on the TCDS). (p. 2-15)
- **Major** = could appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other airworthiness qualities (or not done by elementary operations). **Minor** = everything else. (p. 2-15; Part 43 Appendix A)

Job type	Data required	Paperwork
Minor repair/alteration	ACCEPTABLE data (manufacturer manual, AC 43.13-1)	Ordinary logbook entry (43.9)
Major repair/alteration	APPROVED data (AD, STC, TCDS, DER approval, FAA-approved SB)	FAA Form 337

Field Approval: if there's no existing approved data, the mechanic fills the front of Form 337 through block 6 (leaving block 3 open), describes the work in block 8, and sends it to the local **FSDO** (Flight Standards District Office) for an **ASI** (Aviation Safety Inspector) to approve in block 3. (pp. 2-15, 2-16)

Preventive maintenance - what it IS: Per Part 43, Appendix A paragraph (c), preventive maintenance is **simple/minor preservation operations and replacement of small standard parts NOT involving complex assembly operations**. A holder of at least a **private pilot certificate** may do preventive maintenance on an aircraft they **own or operate**, as long as it is NOT operated under Part 121, 129, or 135. (A **sport pilot** may do preventive maintenance only on a light-sport aircraft they own/operate.) The pilot must still make a record entry. (pp. 2-14, 2-16, 2-20)

Part 43 Appendices (recognize what each is)

- **Appendix A** - lists Major Alterations, Major Repairs, and Preventive Maintenance. (p. 2-19)
- **Appendix B** - Recording of Major Repairs/Alterations (the **Form 337** rules). (pp. 2-19, 2-20)
- **Appendix C - Reserved** (empty). (p. 2-20)
- **Appendix D** - Scope & Detail of **Annual and 100-hour inspections** (the list is the SAME for both; only who-can-sign differs). **8 detailed areas**. (p. 2-20)
- **Appendix E - Altimeter system test** ("the 411 test") - every 24 calendar months for IFR. (p. 2-20)
- **Appendix F - ATC transponder test** ("the 413 test") - every 24 calendar months. (p. 2-20)

MUST KNOW - Section 7 (Part 43)

- **43.3** = who may **PERFORM** (9 persons). **43.7** = who may **APPROVE** for RTS (7 persons). A repairman may **PERFORM** but **NOT** approve RTS.
- **43.9** = maintenance entries; **43.11** = inspection entries. Both = **What (work done) / When (date) / Who (signature, certificate number, type of certificate)**.
- **OVERHAUL** = tested to service limits; **REBUILT** = tested to **NEW**-part tolerances.

- **Minor = acceptable data + logbook entry. Major = approved data + FAA Form 337.**
- **Repair** returns to previous condition; **Alteration** changes it. **Major** = appreciably affects weight/balance/strength/performance/etc.
- **411 test (Appendix E) = altimeter, every 24 months. 413 test (Appendix F) = transponder, every 24 months.** Appendix C = reserved. Annual and 100-hour use the SAME Appendix D list.

8. Inside Part 91, Subpart E - Inspections & the Record-Keeping Rules

Section	What it does
91.403	Owner/operator is primarily responsible for keeping the aircraft airworthy, including AD compliance. (p. 2-21)
91.405	Owner must get required inspections done, fix discrepancies per Part 43, and ensure entries are made; placard inoperative equipment. (p. 2-21)
91.407	Operation after maintenance - needs RTS + entry; an operational flight check is required if the work could have appreciably changed flight characteristics. (p. 2-21)
91.409	Inspections. Annual inspection required; if used for hire, also 100-hour . A 100-hour may be exceeded by up to 10 hours only to fly to the inspection facility (and that overage is subtracted from the next interval). Covers progressive inspection too. (pp. 2-21, 2-22)
91.411	"411 test" - altimeter / static system / altitude reporting, every 24 calendar months . (p. 2-22)
91.413	"413 test" - ATC transponder, every 24 calendar months . (p. 2-22)
91.417	MAINTENANCE RECORDS - the temporary vs permanent split (Section 9 below). (pp. 2-22, 2-23)
91.419	Transfer of records - records go with the aircraft when sold (may be plain language or coded). (p. 2-23)
91.421	Rebuilt engine records - may get "ZERO TIME" (a new record, no previous operating history) when rebuilt by the manufacturer or an approved agency. (p. 2-23)

Annual vs. 100-hour - the key idea: the **inspection itself is the same** (same Appendix D list). The difference is **who can sign**: an **annual must be performed by an IA (Inspection Authorization holder)**; a 100-hour can be signed by an A&P. (Per 14 CFR 65.95(a)(2).) (pp. 2-20, 2-22)

9. Record-Keeping: TEMPORARY vs PERMANENT (91.417) - the most-tested split

This is the single biggest exam topic in the chapter. **14 CFR 91.417 = how long records are kept**. (Compare: **43.9 / 43.11** = how to *make* the entry; **91.417** = how long to *keep* it.) (pp. 2-22, 2-23, 2-39, 2-40)

	TEMPORARY records 91.417(a)(1) & (b)(1)	PERMANENT records 91.417(a)(2) & (b)(2)
What they are	Records of maintenance, preventive maintenance, alterations, and ALL inspections - the routine "what/when/who" entries	The big-history records: total time in service (airframe, each engine, each propeller, each rotor); current status of life-limited parts ; time since last overhaul of items required to be overhauled; current inspection status ; current status of applicable ADs ; copies of Form 337 for major alterations
How long kept	1 year from when the work was done - OR until the work	Kept as long as you operate the aircraft , and TRANSFERRED WITH THE AIRCRAFT when it is sold

	is repeated or superseded, whichever comes first	
Memory hook	"Temporary = the everyday entries that get replaced next time" (1 year / until superseded)	"Permanent = the aircraft's life story that travels with it forever"

Two more 91.417 pieces (don't miss these):

- **91.417(a)** (the 411 & 413 testing records): kept **2 years** (because those tests are every 24 months). *(p. 2-22, Fig. 2-10)*
- **91.417(d)**: the **Form 337 for extended-range fuel tanks** installed in the cabin/cargo area must be **kept ON BOARD the aircraft at all times**. *(pp. 2-22, 2-20)*
- **91.417(b)(3)**: the list of **defects/discrepancies** given to the owner (per 43.11) is kept until those defects are repaired and the aircraft is approved for RTS. *(Fig. 2-10, p. 2-22)*

Where records live: Other than the 91.417(d) fuel-tank Form 337, maintenance records do NOT have to be carried on the aircraft - it's actually safer to keep them in an office and back them up (scan/copy) off-site against fire/flood. Records must be made available to the FAA or NTSB on request (91.417(c)). *(pp. 2-22, 2-23)*

MUST KNOW - Section 9 (the 91.417 split - memorize cold)

- **TEMPORARY = 91.417(a)(1) & (b)(1)** = maintenance, preventive maint., alterations, & all inspections = kept **1 YEAR or until the work is repeated/superseded**.
- **PERMANENT = 91.417(a)(2) & (b)(2)** = total time in service, life-limited parts status, time since overhaul, current inspection status, current AD status, Form 337s = **kept while you own it and TRANSFERRED WITH THE AIRCRAFT**.
- **91.417(a)** (411/413 test records) = kept **2 years**. **91.417(d)** = fuel-tank Form 337 stays **ON the aircraft**.
- **43.9 / 43.11 = how to make the entry; 91.417 = how long to keep it**.
- **Rebuilt engine -> "ZERO TIME"** record (91.421).

10. The FAA Document Types (what each one is FOR)

Document	Full name	What it's FOR / key fact
AC	Advisory Circular	GUIDANCE, not binding (unless referenced into a reg). Shows an "acceptable" way to comply. Numbered to match the reg subject (e.g., 43.13 relates to Part 43.13). AC system started 1962 . (pp. 2-24, 2-25)
AD	Airworthiness Directive	MANDATORY (it IS a reg - Part 39). Issued for an unsafe condition likely to exist in other products of the same design . Printed in the Federal Register, "incorporated by reference" into Part 39. You may operate the product ONLY after the AD is complied with. (pp. 2-8, 2-26)
SAIB	Special Airworthiness Information Bulletin	Information only - NOT mandatory . Alerts/educates about a concern that does NOT rise to the level of an AD. (p. 2-27)
TCDS	Type Certificate Data Sheet	The aircraft/engine/prop's " spec sheet " - lists limits & conditions for type certification (airspeed, weight, thrust, engine model, fuel type, min crew, etc.). It is PART OF the TC (per 21.41) and is evidence the product was type certificated. First published Jan 1958 . (pp. 2-30, 2-31)
STC	Supplemental Type Certificate	FAA approval of a modification to a type-certificated product. Supplements the original TC; defines the design change, how it affects type design, and serial-number effectivity. The STC & its data are the property of the STC holder - you must get permission to use it. (pp. 2-27, 2-30)
TC	Type Certificate	The FAA's approval of an aircraft/engine/prop design (its "birth certificate"). May only be changed by FAA authorization (that change = an STC). (pp. 2-27, 2-30)
Aircraft Specifications	(older recordkeeping document)	FAA recordkeeping documents (from the 1926 Air Commerce Act era) for older products. No longer issued but still in effect. Can be converted to a TCDS at the TC holder's option. Subdivided into 5 groups . A specification is NOT part of a TC . (p. 2-27)

How AC and AD numbers are built (recognize the pattern)

- **AC number** (e.g., 25-42-C): first part = the **FAR subject/Part** (25); middle = **sequential** number within that subject (42nd); last letter = **revision** (C = 3rd revision; first issue has no letter). (p. 2-26)
- **AD number** (3 parts): **(1) year** of issue, **(2) the biweekly period** of the year, **(3) sequential** number within that biweekly period. (p. 2-26)

Three types of AD

- **NPRM** (Notice of Proposed Rulemaking) followed by a Final Rule - the **standard** process (public comment first). (p. 2-26)
- **Final Rule; Request for Comments** - urgent enough to adopt immediately (comment period < 60 days), comments taken after. (p. 2-26)

- **Emergency AD** - immediate action needed; sent by fax/letter; effective by "**actual notice**" only to those who receive it; a matching Final Rule AD is published in the Federal Register (usually within 30 days). (p. 2-26)

AMOC = Alternative Method of Compliance: a different technique (or compliance time) than the AD specifies, allowed AFTER FAA approval if it gives an equal level of safety. (p. 2-27)

Non-FAA documents

- **ATA iSpec 2200** (formerly ATA 100) - the airline-industry system that gives **standard chapter numbers** to aircraft systems so technical data is consistent across large aircraft (e.g., Ch 28 = Fuel, Ch 32 = Landing Gear, Ch 24 = Electrical Power). (pp. 2-30, 2-38)
- **Service Bulletin (SB)** - the manufacturer's notice about a defect, service change, or improvement. **NOT mandatory** unless called out by an AD or required by the inspection program. (Ref 39.27.) (p. 2-33)
- **Structural Repair Manual (SRM)** - detailed structural repair limits/criteria; usually a separate manual on large aircraft, often folded into the AMM on small ones. (p. 2-33)
- **Airworthiness Limitations** - a separate, clearly titled section of the ICA listing **mandatory replacement times, structural inspection intervals, and related procedures**. (p. 2-33)
- Manufacturer manuals: **AFM** (Airplane Flight Manual - serial-number specific) / **POH** (Pilot's Operating Handbook - model specific); **AMM** (Aircraft Maintenance Manual - whole aircraft) / **CMM** (Component Maintenance Manual - one component, often the overhaul manual). (pp. 2-33, 2-34)

MUST KNOW - Section 10

- **AC = advisory (not mandatory). AD = mandatory (Part 39, unsafe condition likely in same-design products).**
- **SAIB = information only, not mandatory** (concern below AD level). **SB = not mandatory** unless an AD or inspection program requires it.
- **TCDS = the spec sheet; it IS part of the TC. STC = approval of a MODIFICATION, supplements the TC. TC = design approval.**
- **3 AD types:** NPRM->Final Rule (standard); Final Rule w/ Request for Comments; **Emergency AD** ("actual notice").
- **AMOC = Alternative Method of Compliance** (needs FAA approval).
- **ATA iSpec 2200 = standard chapter numbers for aircraft systems.**

11. The FORMS (and the Two Certificates That Live in the Aircraft)

Form / Certificate	What it is & key facts
Airworthiness Certificate	Shows the aircraft is airworthy. Two kinds: STANDARD = FAA Form 8100-2 (Normal, Utility, Acrobatic, Commuter, Transport, Manned free balloon, Special classes); SPECIAL = FAA Form 8130-7 (Primary, Restricted, Multiple, Limited, Light-sport, Experimental, Special flight permit, Provisional - the experimental cert is PINK). Must be ON BOARD and DISPLAYED at the cabin/flight-deck entrance, legible to passengers/crew (91.203). (pp. 2-34, 2-35)
Aircraft Registration	Shows ownership . Applied for on FAA Form 8050-1 ; approval issues AC Form 8050-3, Certificate of Aircraft Registration . A copy of the application is temporary authority to operate for up to 90 days . Either the pink application copy or the actual certificate must be on board (91.203(a)(2)). (pp. 2-37, 2-38)
Radio Station License	An FCC (not FAA) document. Required ONLY if the aircraft flies OUTSIDE the U.S. Not required for domestic-only operation (eliminated for domestic use in 1996). (p. 2-38)
FAA Form 337	Major Repair and Alteration form. Required whenever a major repair or major alteration is done (the one exception: a Part 145 repair station may use a signed work order + maintenance release instead). Completed at least in duplicate - original to the owner , and a copy goes to the FAA. (Since fall 2005, the technician sends the copy directly to the Aircraft Registration Branch, Oklahoma City ; historically a copy went to the local FSDO within 48 hours .) Extended-range fuel tanks in cabin/cargo require a THIRD copy kept on board the aircraft . Item 8 (back side) = description of work + the approved-data reference. (pp. 2-20, 2-38, 2-39)

MUST KNOW - Section 11 (forms - memorize the form numbers)

- **Standard Airworthiness Cert = FAA Form 8100-2. Special Airworthiness Cert = FAA Form 8130-7** (experimental = PINK).
- **Registration applied for on Form 8050-1; certificate issued = AC Form 8050-3.** Application = 90 days temporary authority.
- **FAA Form 337 = Major Repair and Alteration.** Original to owner; copy to FAA (now Oklahoma City Registration Branch directly). Extra fuel-tank copy stays on board.
- **Radio Station License = FCC, required only for flights OUTSIDE the U.S.**
- Airworthiness certificate must be **on board AND displayed** at the cabin/flight-deck entrance.

12. Records: Making Entries & Electronic Records

- **43.9 / 43.11** say you **MUST** make a maintenance/inspection entry; **91.417** says how long to keep it (see Section 9). (p. 2-39)
- **Electronic records** are allowed and now common; programs track ADs, SBs, STCs, TCDs, times in service, AD compliance, etc. The "record" definition in Part 3 already includes electronic format. The technician should know the benefits/limits of these systems. (pp. 2-1, 2-39, 2-40)

13. Light Sport Aircraft (LSA) Maintenance - the Basics

The **LSA** category (added in 2004) includes gliders, airplanes, gyroplanes, powered parachutes, weight-shift, and lighter-than-air. Two types: **SLSA = Special (FACTORY-built)** and **ELSA = Experimental (KIT-built)**. (p. 2-40)

How LSA maintenance is different from regular aircraft maintenance (this is the testable contrast):

- LSA rules come from an **industry consensus standard (ASTM)**, not the usual FAA manuals. The relevant practice is **ASTM F2483**. (pp. 2-40, 2-41, 2-44)
- **FAA Form 337 is NOT used** for LSA major repairs/alterations. Instead the work is evaluated against the ASTM standard and a **written affidavit** is provided that the LSA still meets the ASTM requirements. (p. 2-45)
- **Safety directives** (not ADs) are issued for LSA by the **manufacturer**, not the FAA. They are **mandatory - EXCEPT for experimental-use (ELSA) aircraft**. (But if the LSA contains an FAA-TC'd product, the manufacturer must issue a safety directive.) (pp. 2-45, 2-46)
- **Who can work on an LSA** (by maintenance level): (pp. 2-44, 2-45)
 - **Line maintenance** (servicing, simple R&R, the 100-hour/annual condition inspection) - minimum cert = **LSA Repairman with INSPECTION rating** (may inspect an aircraft they own).
 - **Heavy maintenance** - a certified mechanic (A and/or P) OR an **LSA Repairman with MAINTENANCE rating** who has extra task-specific training (this rating can do maintenance AND the 100-hour/annual).
 - **Overhaul** - only the manufacturer (or someone they authorize); requires a separate overhaul manual.
- **Annual condition inspection** = a detailed inspection done once a year per the LSA's maintenance manual. The **100-hour inspection is the same as the annual, just a different interval**. (pp. 2-44, 2-45)

MUST KNOW - Section 13 (LSA)

- **SLSA = Special = factory-built. ELSA = Experimental = kit-built**. Category added 2004.
- **LSA uses ASTM consensus standards (F2483), NOT Form 337**. Major LSA work -> written **affidavit** of ASTM compliance instead of a 337.
- **LSA gets manufacturer "Safety Directives," not FAA ADs** - mandatory except on experimental (ELSA) aircraft.
- **LSA Repairman ratings: INSPECTION** (line maint. + 100-hr/annual on aircraft they own) vs **MAINTENANCE** (maint. + 100-hr/annual, with task-specific training). **Overhaul = manufacturer only**.

14. The 12 Highest-Yield Facts (your last-minute review)

If you only re-read one box before class, make it this one

1. **Airworthy** = conforms to type design + in condition for safe operation. (Part 3.5)
2. **The 3 big regs:** Part 21 (certification), Part 43 (maintenance), Part 91 (operating).
3. **AC = advisory (not binding). AD = mandatory (Part 39).** Regulation = law; guidance = advice.
4. **FAA Form 337 = Major Repair and Alteration.** Original to owner; copy to FAA (Oklahoma City).
5. **Minor = acceptable data + logbook. Major = approved data + Form 337.**
6. **43.9 = maintenance entry; 43.11 = inspection entry** (both What / When / Who).
7. **91.417 TEMPORARY (a)(1)/(b)(1):** maint., prev. maint., alterations, all inspections - kept **1 yr / until repeated or superseded.**
8. **91.417 PERMANENT (a)(2)/(b)(2):** total time in service, life-limited parts, time since overhaul, current inspection status, current AD status, Form 337s - **kept while owned & TRANSFERRED WITH THE AIRCRAFT.**
9. **OVERHAUL = service limits; REBUILT = new-part tolerances.** Rebuilt engine can get "**zero time.**"
10. **TCDS** = spec sheet, part of the TC. **STC** = approval of a modification. **SAIB** = info only.
11. **Standard A/W cert = 8100-2; Special A/W cert = 8130-7** (experimental = pink). Must be displayed in aircraft.
12. **SLSA = factory-built; ELSA = kit-built. LSA uses ASTM (not 337) and manufacturer Safety Directives (not ADs).**

Built by amt-tutor from FAA-H-8083-30(B) Chapter 2 (instructor-assigned PDF). No quiz or flashcards included by request - this is a read-through summary only. When in-class slides arrive, the worksheet/written split gets checked against AIM wording.

Block 3 — Chapter 13: Mechanic Privileges & Limitations

Plain-English Study Guide — read this instead of slogging the chapter, still know everything

Block 3 — General | Built 2026-06-14 | Source of truth: assigned chapter PDF "*Chapter 13 Mechanic Privileges and Limitations*" (8 pp.), built on **14 CFR Part 65** — Subpart A (General), Subpart D (Mechanics), and the Inspection Authorization sections. | **This chapter is reg-number heavy — the section numbers and timeframes get tested COLD. Memorize them.** | SUMMARY ONLY (no quiz/flashcards yet, per your request).

0. The Big Picture (read this first)

What Part 65 is: Part 65 is the FAA rulebook for certifying **airmen who are NOT flight crew** — i.e., the people on the ground who keep airplanes legal and safe. ("Airman" here just means an FAA-certified person, not necessarily a pilot.) Part 65 has six subparts; this chapter only cares about **two of them: Subpart A = General rules** (apply to everybody) and **Subpart D = Mechanics** (the A&P rules), plus the **Inspection Authorization (IA)** sections. (p. 1)

Two kinds of maintenance technician — know the difference cold:

- **Mechanic certificate (the A&P) = transportable.** It belongs to YOU based on your training and knowledge, and it travels with you no matter where you work.
- **Repairman certificate = tied to one location/employer.** It is issued to you only while you work at a **specific company at a specific address**, and that address is printed on it. When you leave that job, you must **return the repairman certificate to the FSDO** (Flight Standards District Office) that issued it.

Analogy: A mechanic certificate is like a **driver's license** — it's yours, it works anywhere. A repairman certificate is like a **building access badge** — it only works at the one company that issued it, and you hand it back at the door when you quit. (p. 1)

1. Subpart A — General Rules (the §65.1x block)

These apply to **all** Part 65 airmen, not just mechanics. They are the "paperwork, honesty, and testing" rules. The full reference table is below; the high-yield numbers are spelled out underneath it.

14 CFR §	Title (plain meaning)
SUBPART A — GENERAL (applies to all Part 65 certificate holders) — ch. pp. 1–3	
65.3	Certification of foreign airmen. FAA normally issues these certificates only to U.S. citizens or resident aliens . Exception: it will issue to a person OUTSIDE the U.S. if needed for the operation/continued airworthiness of a U.S.-registered civil aircraft , if they meet the requirements.
65.11	

	Application & issue. Apply on FAA Form 8610-2 (Airman Certificate and/or Rating Application). If SUSPENDED , you may not apply for additional ratings during the suspension. If REVOKED , you may not apply for a certificate for 1 year after the revocation.
65.12	Offenses involving alcohol & drugs. A conviction for a drug offense (growing, processing, manufacturing, selling, disposing, possessing, transporting, or importing narcotics/marijuana/depressants/stimulants) can get your application denied for up to 1 year after the conviction date, and can cause suspension or revocation of a certificate you already hold.
65.13	Temporary certificate. Pass all required tests with at least 70% and you may get a temporary certificate good for not more than 120 days , while the FAA reviews your paperwork and issues the permanent one.
65.14	Security disqualification. Added after 9/11/2001 . If TSA determines you are a security threat , your application is held, or a certificate you hold is revoked .
65.15	Duration of certificates. A mechanic certificate is effective until surrendered, suspended, or revoked (no expiration date). Surrendered = you give it up voluntarily. Suspended = FAA temporarily removes it. Revoked = FAA permanently removes it.
65.16	Change of name / replacement of lost or destroyed certificate. A name change must include your current certificate plus a marriage license, court order, or other verifying document . Name-change and replacement requests go to the FAA Airmen Certification Branch (AFB-720), P.O. Box 25082, Oklahoma City, OK 73125 . There is a nominal charge .
65.17	Test: general procedure. Tests are given by FAA-designated persons. Minimum passing score = 70%.
65.18	Written tests: cheating / unauthorized conduct. If you cheat (copy/remove the test, give or receive any part of it or any help, take it for someone else, or use unauthorized aids), you are ineligible for any certificate or rating for 1 year , and current ratings may be suspended or revoked .
65.19	Retesting after failure. Two options: (1) wait 30 days from the failure date, then retest; OR (2) get additional instruction and bring a signed statement from the certificated technician who instructed you saying you're ready — then retest with no wait .
65.20	Falsification, reproduction, or alteration. Grounds to suspend or revoke any certificate/rating: a fraudulent or intentionally false statement on an application ; a fraudulent/false entry in any logbook, record, or report required for a certificate; reproducing a certificate for fraud; or altering a certificate. (Ties to part 43 §§43.9 & 43.11 record-entry rules.)
65.21	Change of address. Notify the FAA in writing within 30 days of a change of permanent residence (same Oklahoma City Airmen Certification Branch address).

Plain-words drill-down on the heavily-tested General sections

The three “states” of a certificate (§65.15) — the FAA loves to test these definitions:

- **SURRENDERED** = you gave it up **voluntarily**.
- **SUSPENDED** = FAA pulled it **temporarily** (you can get it back). After a suspension → you can't apply for added ratings *during* the suspension (§65.11).

- **REVOKED** = FAA pulled it **permanently**. After a revocation → you can't even apply again for **1 year** (§65.11).

The two retesting paths (§65.19): Think of it as “wait or weight.” Either **WAIT 30 days**, or add the **WEIGHT** of a signed instructor statement and retest immediately. Those are the only two paths.

(p. 2)

Bonus topic the chapter folds in here (drug/alcohol testing — part 120, not part 65): Refusing a required drug/alcohol test (under **14 CFR part 120**) can get your application denied and existing certificates suspended/revoked. **§120.117** (drug program) requires a **urine sample**; **§120.37** (misuse of alcohol) requires a **breath test**. Memorize the **seven times testing can be required**: (1) **pre-employment**, (2) **periodic**, (3) **random**, (4) **post-accident**, (5) **reasonable cause**, (6) **return-to-duty**, (7) **follow-up**. (pp. 2–3)

2. Subpart D — Mechanics (the §65.7x & §65.8x block)

This is the heart of the chapter — the rules for **becoming** an A&P and what an A&P **may and may not do**.

14 CFR §	Title (plain meaning)
SUBPART D — MECHANICS (the A&P rules) — ch. pp. 3–5	
65.71	Eligibility (general). Must be 18+ years old ; read/write/speak/understand English ; have passed all required tests within the preceding 24 months ; and possess/demonstrate the right knowledge & skill for the rating sought.
65.73	Ratings. Two ratings: Airframe and Powerplant . Can be earned separately or as a combined A&P. Old A or E certificates valid before June 15, 1952 can be exchanged for the current equivalents (A & E → A&P).
65.75	Knowledge (written) requirements. Pass written test(s) at 70% (§65.17). Three tests: General = 60 questions, Airframe = 100, Powerplant = 100. Includes applicable parts of 14 CFR 43 & 91 ; propeller install/maintenance basics are in the Powerplant test. Must pass written BEFORE applying for the oral & practical (§65.79).
65.77	Experience requirements. Either a certificate of completion from a part 147 certificated AMTS (your AIM school), OR documented practical experience: 18 months for ONE rating (airframe OR powerplant), 30 months for BOTH.
65.79	Skill (oral & practical) requirements. Take the oral and practical tests AFTER passing the written . Powerplant practical also requires demonstrating minor repairs & minor alterations to propellers . Standards are in the Aviation Mechanic Practical Test Standards (PTS) .
65.80	Certificated AMTS students. A part 147 school student making satisfactory progress may take the oral & practical (§65.79) BEFORE finishing the curriculum and BEFORE taking the written (§65.75).
65.81	General privileges & limitations. See the “may / may NOT” box below — this one is tested hard.
65.83	Recent experience requirements. Can't exercise the certificate unless you've been actively working as a mechanic at least 6 of the preceding 24 months (see box below).
65.85	

	Airframe rating — additional privileges. May approve & return to service the airframe/appliance/part after minor repairs/alterations, and may do the work for a major repair/alteration (start FAA Form 337) — but the return to service for a major must be signed by an IA (§65.95). May do the airframe 100-hour inspection (per part 91 §91.409). Light-sport exception noted below.
65.87	Powerplant rating — additional privileges. Same deal for powerplant & propeller: return to service minor repairs/alterations; do the work for a major but the return to service must be signed by an IA ; may do the 100-hour inspection on a powerplant/propeller. Light-sport exception noted below.
65.89	Display of certificate. Keep the certificate in the immediate area where you normally work, and present it on request to the FAA, NTSB, or any federal/state/local law-enforcement officer.

§65.71 — Eligibility, spelled out (memorize the four)

To get a mechanic certificate you must:

1. Be at least **18 years old**.
2. **Read, write, speak, and understand English.** (If you can't and you work outside the U.S. for a U.S. carrier, the cert is endorsed “**valid only outside the United States.**”)
3. Have **passed all required tests (written, oral, practical) within the preceding 24 months.**
4. **Possess and demonstrate** the appropriate knowledge and skill for the rating.

Adding the second rating later? Same deal — meet §65.77 experience and take written+oral+practical **within 24 months**. Memory hook: “**18 / English / 24 months / know-and-show.**” (p. 3)

§65.81 — General Privileges & Limitations (the “may / may NOT” rule)

An A&P **MAY**: perform or supervise the maintenance, preventive maintenance, or alterations of an aircraft or appliance (or part) **for which they are rated**. The mechanic must **have and understand the maintenance manuals / instructions for continued airworthiness** for the task.

An A&P may **NOT**:

- Do a **major repair or major alteration to a PROPELLER.**
- Do **ANY** repair to, or alteration of, **INSTRUMENTS.** (Props-major and instruments are reserved for **certificated repairmen at an authorized repair station.**)
- **Supervise** a task **unless they have satisfactorily performed that same work at an earlier date.**

Why the OJT log matters: Because of that last rule, you can only **supervise** work you've already **done yourself** and logged. That's why keeping an **On-the-Job-Training (OJT) log** is huge — it's

your proof. **Memory hook for the two big NO's: "No major PROP, no INSTRUMENTS"** — PI (Props/Instruments) are off-limits. (p. 4)

§65.83 — Recent Experience (the "6 of 24" rule)

Your A&P never **expires** (§65.15) — but you can't **USE** it unless you've been active as a mechanic for **at least 6 of the preceding 24 months**. "Active" counts if you did **any one (or a mix)** of:

- Served as a **mechanic** under the certificate and rating;
- **Technically supervised** other mechanics;
- **Supervised, in an executive capacity**, the maintenance or alteration of an aircraft.

Memory hook: "6 of 24 keeps it alive." Don't confuse the certificate **existing** (forever, until surrendered/suspended/revoked) with the certificate being **exercisable** (only if 6-of-24 is met). The FAA loves that trap. (p. 4)

§65.85 / §65.87 light-sport exception (know it exists): An airframe (or powerplant) mechanic **can** approve & return to service a **major** repair/alteration on a **special airworthiness certificate light-sport aircraft** (per part 21 §21.190) **without an IA** — but only on products **not produced under FAA approval (not type certificated)** and done per manufacturer (or FAA-acceptable) instructions. For normal type-certificated aircraft, a **major** still needs IA sign-off. (p. 4)

3. Inspection Authorization — IA (the §65.9x block)

The IA is the **next step up** from A&P. It lets you **approve major repairs/alterations for return to service** and perform **annual inspections** — things a plain A&P can't finish alone. These four sections are **heavily tested**.

14 CFR §	Title (plain meaning)
INSPECTION AUTHORIZATION (IA) — ch. pp. 5–6	
65.91	IA — eligibility. Hold an A&P for at least 3 years, active the last 2 years ; apply on FAA Form 8610-1 ; have a fixed base of operation + equipment, facilities, and inspection data ; then pass a written test at 70% . Fail → wait 90 days to retest (no reduction for extra training).
65.92	IA — duration. Expires March 31 of each ODD-numbered year ; only usable while you hold a current mechanic certificate. Ceases if surrendered/suspended/revoked, OR you lose your fixed base, OR you lose your facilities/equipment/inspection data.
65.93	IA — renewal. Each year, renew by meeting any one of the activity options below (see box). Held the IA <90 days → no renewal required that cycle.
65.95	IA — privileges & limitations. May do an annual inspection; perform/supervise a progressive inspection; and approve for return to service after a major repair or alteration — EXCEPT aircraft on a continuous-airworthiness program under part 121 . Keep certificate available for: aircraft owner, A&P, FAA, NTSB, or any law-enforcement officer. If you move your fixed base , notify the new FSDO in writing BEFORE exercising IA privileges there.

§65.91 — Getting the IA, spelled out

The “3 and 2” rule: hold the A&P **3 years**, and be **active the last 2 years**. Plus: a **fixed base of operation** (a place you can be reached in person or by phone during a normal work week — it does NOT have to be where you actually do inspections); the **equipment, facilities, and inspection data** to properly inspect; apply on **Form 8610-1**; then **pass the written at 70%**. **Fail = 90-day wait** before retesting, and (unlike the A&P test) **extra training does NOT shorten the wait**. (p. 5)

§65.93 — Renewing the IA

Each year, do **ANY ONE** of these (you cannot **mix** them):

- **One annual inspection per 90 days** you’ve held the IA (so **4 annuals** over the year);
- **Two major repair/alteration inspections per 90 days** (so **8 over the year** — count whether approved or disapproved);
- Perform/supervise & approve **one progressive inspection**;
- Complete a **refresher course of at least 8 hours** acceptable to the Administrator;
- Pass an **oral test** by an FAA inspector (the “last resort” method).

The **cumulative trick the chapter calls out**: the **4 annuals / 8 major-inspections** are **cumulative by March 31**, not required in each individual 90-day chunk. You could go 11 months doing nothing, then do all 4 annuals (or all 8 337-inspections) in March. But you **cannot mix** methods (e.g., 2 annuals + 4 Form 337s does not count). (pp. 5–6)

⚠ **MUST KNOW — memorize these numbers & facts COLD**

- **70%** = minimum passing score on every test (§65.13, §65.17, §65.75, §65.91).
- **120 days** = max validity of a **temporary certificate** (§65.13).
- **1 year** = denial window after a **drug conviction** (§65.12), a **cheating** finding (§65.18), or a **revocation** before you can re-apply (§65.11).
- **30 days** = retest wait after a written failure (§65.19) *and* the window to report a **change of address** (§65.21).
- **18 years old + English + tests within 24 months** = A&P eligibility (§65.71).
- **Experience: 18 months** for ONE rating, **30 months** for BOTH (§65.77).
- **Written question counts**: General **60**, Airframe **100**, Powerplant **100** (§65.75).
- **§65.81 limits**: NO major repair/alteration to **PROPELLERS**; NO repair/alteration of **INSTRUMENTS**; can’t supervise work you haven’t done before.
- **§65.83**: active **6 of the preceding 24 months** to **exercise** the certificate (cert itself never expires — §65.15).
- **IA eligibility (§65.91)**: A&P **3 years**, active last **2 years**, Form **8610-1**, written at 70%; **fail** → **90-day wait**.

- **IA expires March 31 of each ODD-numbered year** (§65.92).
- **IA renewal (§65.93): 4 annuals OR 8 major-repair inspections OR 1 progressive OR 8-hr refresher OR oral test — cannot mix**; held <90 days = no renewal that cycle.
- **IA can't approve return to service on part 121 continuous-airworthiness aircraft** (§65.95).
- **Forms: 8610-2** = mechanic application (§65.11); **8610-1** = IA application (§65.91); **Form 337** = major repair & alteration.
- **Cert states:** Surrendered = voluntary; Suspended = temporary; Revoked = permanent (§65.15).

4. Ethics (the closing discussion)

What it is: “Ethics” comes from the Greek “**ethos**” (character / custom) and means “**the study of standards of conduct and moral judgment.**” For an A&P, ethics is doing the right, legal, airworthy thing even when the boss, the schedule, or the bottom line pushes you to cut corners. (p. 6)

The ice-shield scenario (the lesson): A tech removed a damaged fuselage ice shield believing a replacement was in stock; inventory was wrong. **Key rule:** once a damaged part is removed, it becomes **unairworthy and CANNOT be reinstalled** — even if it's deferrable on the **MEL (Minimum Equipment List)**. The inspector told him to reinstall it, then to do an unauthorized epoxy patch the shop wasn't authorized to make. The tech **refused and walked away**; someone else compromised standards and the aircraft **departed illegally and unairworthy**. Lesson: **it's the mechanic's responsibility to follow the regs and question a supervisor who circumvents them for an on-time departure.** (pp. 6–7)

The B.F. Goodrich brake scandal (the second example): A classic **whistle-blowing** case — a brake system that failed testing **14 times** was pushed toward qualification “no matter what,” under a culture of intimidation, leading to a **1969 congressional hearing**. Point: a **company culture of intimidation/distrust** transcends every industry, and **trust** is the foundation of long-term success. (pp. 7–8)

Two checklists worth remembering:

Seven-step ethical-decision checklist: (1) recognize/clarify the dilemma, (2) get all the facts, (3) list ALL options, (4) test each (Is it legal? Is it right? Is it beneficial?), (5) decide, (6) double-check (How would I feel if family knew? if it hit the newspaper?), (7) take action.

Four common ways people rationalize bad behavior: pretending it isn't unethical/illegal; claiming it's in the org's best interest; assuming no one will find out; expecting superiors to protect you (— they won't; in court they'll swear they always told you to work “by the book”). The chapter closes with the **PAMA Code of Ethics** — never approve work beyond your knowledge, and never let money or a superior push you to pass doubtful work as airworthy. (pp. 8)

Block 3 — Chapter 9: Fluid Lines & Fittings

Plain-English Study Guide — read this instead of the whole chapter

amt-tutor make-up study guide · 2026-06-14 · 5th-grade-teacher level · feeds class + cumulative Block exam
· supports hangar projects G22/G23

Source (page numbers cited on every fact): FAA-H-8083-30 (Aviation Maintenance Technician — General), Chapter 9 "Fluid Lines & Fittings," pages 9-1 through 9-23. Citations like [\[p. 9-3\]](#) point straight to the page so you can double-check anything.

The one-sentence big idea: Airplanes move fluids (fuel, oil, hydraulic, oxygen, instrument air) through two kinds of plumbing — **rigid metal tubing** for long straight runs that hold still, and **flexible hose** wherever there is movement or vibration. This chapter is how you cut, bend, flare, fit, identify, install, and inspect both. [\[p. 9-1\]](#)

1. The Two Kinds of Plumbing (start here)

Rigid fluid lines = metal tubing. Used for stationary, long, relatively straight runs — fuel, oil, coolant, oxygen, instrument, and hydraulic lines. [\[p. 9-1\]](#)

Flexible hose is used with *moving* parts or anywhere there is a lot of vibration. [\[p. 9-1\]](#)

5th-grade picture: Rigid tubing is like the copper water pipes bolted inside your house walls — they don't move. Flexible hose is like the bendy hose under your kitchen sink that lets the faucet wiggle without snapping the pipe. On a plane, anything that shakes or moves gets the bendy hose; everything that holds still gets the stiff metal pipe.

Replacement rule: a replacement line must be the *same size and material* as the original. All tubing is pressure-tested before first install and is built to take several times normal operating pressure. If a tube bursts or cracks, it is almost always from **excessive vibration, improper installation, or collision/impact damage** — and you must find out *why* it failed, not just replace it. [\[p. 9-1\]](#)

2. RIGID Tubing — Materials & How to Tell Them Apart

Copper was used in the old days; modern aircraft use **aluminum alloy, corrosion-resistant steel (CRES), or titanium**. [\[p. 9-1\]](#)

Tubing materials and what each one is for

Material	Used for (pressure / job)	Notes & fittings
----------	---------------------------	------------------

1100 H14 & 3003 H14 (1/2-hard aluminum)	General-purpose lines, low or negligible pressure — instrument lines, ventilating conduits	The soft, low-pressure aluminums
2024-T3, 5052-O, 6061-T6 aluminum	Low & medium pressure — hydraulic/ pneumatic 1,000–1,500 psi , fuel & oil lines	The stronger aluminums
Steel — CRES 304, CRES 321, or CRES 304-1/8-hard (annealed)	High-pressure hydraulics (3,000 psi or more) — landing gear, flaps, brakes, fire zones, and anywhere foreign-object damage (FOD) is a risk (gear / wheel wells)	Higher strength = thinner walls, so weight isn't much more than aluminum. Uses swaged or MS flareless fittings
Titanium 3AL–2.5V	Transport-category & high-performance hydraulics, above 1,500 psi	30% stronger than steel, 50% lighter than steel. Uses Cryofit or swaged fittings

Exam trap / safety rule: NEVER use titanium tubing or fittings in any oxygen system. Titanium is oxygen-reactive — a fresh titanium surface exposed to gaseous oxygen can **spontaneously catch fire even at low pressure.** [p. 9-1]

Identifying which aluminum alloy you have — the color-code bands

You can spot aluminum vs. steel vs. titanium by eye, but you **cannot** eyeball which *aluminum alloy* it is. [p. 9-1] On **large** aluminum tubing the alloy number is **stamped** on the surface. On **small** tubing it's usually a **color-code band, no more than 4" wide**, painted at the two ends and about midway. If the band is two colors, each color gets half the width. [p. 9-1] If the code is unreadable, you **hardness-test** a sample to identify it. [p. 9-2]

Aluminum Alloy	Color of Band	Aluminum Alloy	Color of Band
1100	White	5052	Purple
3003	Green	6053	Black
2014	Gray	6061	Blue and Yellow
2024	Red	7075	Brown and Yellow

From Figure 9-1, p. 9-2. To positively match a replacement, compare its code markings against the original tubing you're removing. [p. 9-1]

3. Tubing SIZE — the OD rule (memorize this)

Rigid metal tubing is sized by OUTSIDE DIAMETER (OD), measured in *sixteenths of an inch*. [p. 9-2]

So “number 6” tubing = $6/16" = 3/8"$ OD. “Number 8” = $8/16" = 1/2"$ OD. The OD is printed right on the tube.

You also need the **wall thickness** — printed on the tube in *thousandths of an inch*. To get the inside diameter (ID): **ID = OD – (2 × wall thickness)**. Example: a number 10 tube ($10/16" = 0.625"$ OD) with 0.063" walls has ID = $0.625" - 2(0.063") = 0.499"$. [p. 9-2]

Remember the contrast (it's a favorite test pairing): RIGID TUBE = sized by **OD**.
HOSE = sized by **ID** (covered in Section 9). Two different rules — don't mix them up.

4. Fabricating a Metal Tube Line — Cut, Bend, Flare, Bead

Tube forming is **four processes: cutting, bending, flaring, and beading**. Small soft tubing can be hand-bent during install; tubing **1/4" diameter or larger cannot be hand-bent** without tools. [p. 9-2]

4a. Cutting

Goal: a **square end, free of burrs**. Use a tube cutter (good for soft copper/aluminum) or a fine-tooth hacksaw. Special **chipless cutters** exist for hard 6061-T6, CRES steel, and titanium. [p. 9-2]

1. Cut the new piece about **10% longer** than the old one to allow for bending variations. [p. 9-2]
2. Put the tube in the cutter, cutting wheel on the mark; rotate around the tube, adding light pressure by twisting the thumbscrew a little at a time. **Too much pressure deforms the tube or makes heavy burrs**. [p. 9-2]
3. **Deburr** inside and outside with a knife, the cutter's burring edge, or a deburring tool (turn it end-for-end to do both). [p. 9-2]
4. File the end **square and smooth**. If using a hacksaw, prefer **32 teeth per inch** (a saw work-hardens the metal less than re-cutting). [p. 9-2]

Watch out: when deburring, do NOT thin or fracture the end wall. Even slight damage there leads to **fractured or defective flares that won't seal**. [p. 9-2]

4b. Bending — and why kinks/flat spots fail

Goal: a **smooth bend without flattening**. Tubing under 1/4" can usually bend by hand; larger sizes use portable hand benders or production benders. [p. 9-2] Standard bend radii vary with tube size (Figure 9-4). [p. 9-4]

The number to know: a little flattening in a bend is OK, but the flattened (small) diameter **must not be less than 75% of the original OD**. Tubing that is flattened, wrinkled, or has irregular bends **must not be installed**. [p. 9-3]

Why these fail: *Excessive flattening causes fatigue failure* of the tube. *Wrinkled bends* usually come from trying to bend thin-wall tubing without a bender. [p. 9-3] Four bend results to recognize (Figure 9-8): **Perfect** (good), **Flattened** (bad), **Wrinkled** (bad), **Kinked** (bad). [p. 9-3]

Using a hand bender: insert the tube in the groove so the measured end is left of the form block, align the two zeros, line the mark on the tube up with **L** (or **R** if the measured end is on the right), then pull the form handle steadily until the zero lines up with your desired angle on the radius block. Benders are size-specific — pick the right one for the tube diameter (size is stamped on the bender). [p. 9-3]

4c. Alternative bending method (no bender available)

Pack the tube with a **filler** so it can't collapse, then bend it over a forming block. [p. 9-3]

- **Dry-sand method:** plug one end, fill/pack with fine dry sand, plug the other end **tightly** (so plugs don't blow out), then bend over a block shaped to the radius. [p. 9-3]
- **Fusible-alloy method:** fill the tube under hot water with a **fusible alloy that melts at 160 °F**, cool it, bend slowly by hand around a block, then re-melt under hot water to remove the alloy. [p. 9-3]
- Either way: **remove ALL filler particles** and borescope-inspect so nothing gets carried into the system. Never heat the fusible alloy any other way than prescribed, or it sticks inside and ruins the tube. [p. 9-3]

5. FLARING — Single vs. Double (high-yield)

A **flare** is the funnel-shaped, flared-out end of a tube that a fitting clamps against to make a leak-proof seal. Two kinds: the **single flare** and the **double flare**. Flares see very high pressure, so the shape must be exact — **too small** = weak joint that leaks/pulls apart; **too large** = interferes with the

screw threads and leaks; **crooked** = the tube wasn't cut square. You cannot fix a bad flare by adding torque. [p. 9-3 / 9-4]

THE ANGLE TO MEMORIZE: aircraft flaring tools make a **35° to 37°** flare (call it the **37° standard**). **NEVER use an automotive 45° flaring tool** on aircraft tubing. [p. 9-4]

Single flare vs. Double flare — when to use each

	Single Flare	Double Flare
What it is	One flared-out funnel end	The end is belled out, then folded back on itself — two thicknesses
When used	General flaring	Soft aluminum-alloy tubing 3/8" OD and under
Why	Standard	Stops the soft flare from being cut off and failing under pressure; more concentric, seals better , resists the shearing effect of torque

MUST-KNOW flare rule

- **37° is the aircraft flare angle (35°–37°). Automotive 45° is forbidden.**
- **Double flare = soft aluminum alloy, 3/8" OD and under.**

5a. Rolling-type flaring tool instructions

Material limit: rolling-type tools are **only** for soft **copper, aluminum, and brass**. **Do NOT use on corrosion-resistant steel or titanium.** [p. 9-4]

1. Cut the tube squarely and remove all burrs. [p. 9-4]
2. Slip the **fitting nut and sleeve** onto the tube *first* (you can't add them after the flare). [p. 9-4]
3. Loosen the clamp screw, separate the die, and insert the tube in the die hole that matches its size. These tools are **self-gauging** — clamp the tube **flush with the top of the die block** for the correct flare. [p. 9-4 / 9-5]
4. Advance the clamp screw against the end segment, tighten firmly, move the yoke down over the die holder and twist clockwise to lock. [p. 9-5]
5. Turn the feed screw down firmly until you feel **slight resistance** — that means the flare is done. (Always read the tool's own instructions; types vary.) [p. 9-5]

5b. Double-flaring instructions

1. Deburr inside and outside; cut off the end if it looks damaged. [p. 9-4]
2. **Anneal** brass, copper, and aluminum first — heat to a **dull red** and cool rapidly in cold water (this softens the metal so it folds without cracking). [p. 9-4]
3. Open both clamping screws, pick the flaring-bar hole that matches the tube, and set the tube so it extends above the bar by the thickness of the adapter shoulder; tighten the clamps. [p. 9-4]
4. Insert the pilot of the correctly sized **adapter** into the tube, slip the yoke over the bars and center it. [p. 9-4]
5. Advance the cone down until the adapter shoulder rests on the bar — this **bells out** the end. [p. 9-4]
6. Back the cone off, remove the adapter, then advance the cone **directly into the belled end** — this **folds the tube on itself** into a double flare. **Don't overtighten** or you thin the flare wall. [p. 9-4]

5c. Beading (the fourth forming process)

A **bead** is a raised ring rolled around the tube near the end so a hose can slip over it and a clamp can grip without sliding off. Done with a hand beading tool (for **1/4" to 1" OD**), machine rolls, or grip dies; lubricate inside and outside with light oil. Beading machines are usually for large tubing over 1 15/16". [p. 9-7]

6. Fluid Line IDENTIFICATION — color codes, words, symbols

Fluid lines are marked so anyone can instantly read each line's **function, contents, and primary hazard** using **color codes, words, and geometric symbols** (Figure 9-16). [p. 9-7]

How markers are applied:

- Usually **1" tape or decals**. [p. 9-7]
- **Steel tags** instead of tape on lines **4" diameter or larger**, oily-environment lines, hot lines, and some cold lines. [p. 9-7]
- **Paint** in engine compartments — so tapes/decals/tags can't be sucked into the engine induction system. [p. 9-7]
- Markers go on **both ends** of a line, at least once in each compartment a line passes through, and right next to every valve, regulator, filter, or accessory. [p. 9-7]

The three hazard words (memorize what each means):

- **FLAM** = flammable — fuel lines. [p. 9-7]

- **TOXIC** = lines with toxic materials (replaces FLAM). [p. 9-7]
- **PHDAN** = PHysically DANGerous — oxygen, nitrogen, Freon. [p. 9-7]

Who is responsible: the aircraft/engine **manufacturer** does the original markers; the **mechanic** is responsible for **replacing** them when needed. [p. 9-7]

7. Fluid Line END FITTINGS

Fittings join tubing to a unit, to another tube, or to a hose. Two thread types: [p. 9-7]

Pipe threads	Machine threads
Tapered internal & external; the taper itself forms the seal (like household plumbing). Use approved thread lubricant to prevent seizing & leaks — but keep lubricant out of the system.	No taper, no sealing ability (like a nut and bolt). Only pull connections together or attach through bulkheads. Seal is made by a flared connection, crush washer, or synthetic seal . Size given in dash numbers = OD in 16ths of an inch .

Oxygen safety: do NOT use lubricants on oxygen lines — oxygen reacts with petroleum products and can ignite. Special oxygen-safe lubricants exist. [p. 9-7]

Bulkhead fitting: when a line passes through a bulkhead and must be secured to it, the bulkhead-fitting end is **longer** so a locknut can clamp it to the bulkhead. [p. 9-8]

The four fitting types (pressure & material decide which): (1) **bead & clamp** — low/medium pressure only (vacuum, coolant); (2) **AN flared**; (3) **MS flareless**; (4) **permanent** (Permaswage™, Permalite™, Cryofit™). Flared, flareless, and permanent types work at **any pressure**. [p. 9-8]

7a. AN flared fittings

A sleeve + nut; the nut draws the sleeve and the tube's flare tight against a cone-shaped male fitting to seal. The sleeve also supports the tube so vibration doesn't concentrate at the flare edge. [p. 9-8]

Color code: AN **steel** = **black**, AN **aluminum** = **blue**, aluminum-bronze = cadmium-plated/natural. [p. 9-8]

AN flared vs. MS flareless are NOT interchangeable. Tell them apart by the end: **AN flared has a CONE** at the end; **MS flareless has a STRAIGHT end**. [p. 9-9 / 9-8]

8. MS FLARELESS Fittings — the sleeve, presetting & tightening

MS flareless fittings are made for **high-pressure (3,000 psi) hydraulics** with severe vibration or fluctuating pressure. They need **no flaring**. Three parts: a **body**, a **sleeve (ferrule)**, and a **nut**.

[p. 9-9]

How it seals (5th-grade picture): as you tighten the nut, the body squeezes the **sleeve** so its sharp edge **bites into the outside of the tube** like a tiny ring of teeth grabbing the pipe. That bite is the seal. The body's counterbore has a reverse angle (**15° for steel** connectors, **45° for aluminum**) that stops the tube from collapsing inward and adds sealing force. [p. 9-9]

8a. PRESETTING a new flareless assembly (done before installation)

1. Cut the tube to length with **perfectly square ends**; deburr inside and out. Slip on the **nut first, then the sleeve**; lubricate the fitting and nut threads with hydraulic fluid. [p. 9-5/9-6]
2. Put the fitting in a vise. Hold the tube **firmly and squarely, bottomed in the fitting**. Tighten the nut until the sleeve's cutting edge grips the tube — to find that point, **slowly turn the tube back and forth** while tightening; **when the tube no longer turns, the nut is ready for final tightening**. [p. 9-6]
3. **Final tightening (memorize these turn counts):**
 - Aluminum-alloy tubing **up to and including 1/2" OD** → **1 to 1-1/6 turns**.
 - Steel tubing, and aluminum-alloy **over 1/2" OD** → **1-1/6 to 1-1/2 turns**. [p. 9-6]

After presetting, check: the tube must extend **3/32" to 1/8"** beyond the sleeve pilot (or blowoff can occur). Sleeve pilot must contact the tube or have at most **0.005" clearance for aluminum / 0.015" for steel**. Slight tube collapse at the sleeve cut is OK; **no movement of the sleeve pilot except rotation** is allowed. [p. 9-6]

8b. Installing a flareless fitting (in the aircraft)

1. Run the nut down **by hand** until resistance increases. If fingers can't do it, use a wrench but watch for **first signs of bottoming**. [p. 9-12]
2. Final tightening starts where the nut **just begins to bottom**. Then turn the nut **one-sixth turn (one flat on the hex nut)**, holding the connector still with a second wrench. [p. 9-12]
3. Pressure-test. If it leaks, you may tighten an **additional one-sixth turn (one-third turn total max)**. [p. 9-12]
4. Still leaking after one-third turn? **Remove and inspect** for scores, cracks, foreign material, or overtightening damage. [p. 9-12]

CAUTION: Never tighten a flareless nut beyond one-third turn (two flats). That's the maximum before you permanently damage the sleeve and nut. Overtightening drives the

sleeve's cutting edge too deep, weakening the tube so that **normal in-flight vibration can shear it.** [p. 9-14]

9. Other Permanent Fittings (quick hits)

- **Swaged (Permaswage™):** permanent, near maintenance-free; joins hydraulic lines where you won't routinely disconnect. Often used with **titanium and CRES.** Installed with portable hydraulic tooling; to remove, **cut the tube.** [p. 9-9]
- **Permalite™:** mechanically attached by **axial swaging** — a ring is moved along the fitting to deform it into the tube. [p. 9-9]
- **Cryofit:** a standard fitting with a **shape-memory-alloy (Tinel™)** sleeve. Made 3% small, frozen in liquid nitrogen, expanded 5% larger; slipped on, it **warms in 10–15 seconds, shrinks back, and bites the tube** for a permanent seal. Remove only by cutting the tube. Common with **titanium.** [p. 9-9/9-10]

10. Installing & Inspecting RIGID Tubing

Before install: inspect the line, remove dents/scratches, make sure nuts and sleeves are snugly mated and properly flared, and that the line is clean and free of foreign matter. [p. 9-10]

Three install rules the FAA loves to test:

- **Never put compound on the fitting faces or the flare** — it destroys the metal-to-metal contact that makes the seal. [p. 9-10]
- **Align the line BEFORE tightening. Do not pull the assembly into place with torque on the nut.** [p. 9-10]
- **Use a torque wrench and the correct torque.** Overtightening cuts off the flare or ruins the sleeve/nut; undertightening lets the line blow out or leak. [p. 9-10/9-12]

Why bends matter on install: never pick a path with **no bends.** A tube can't be cut/flared accurately enough to fit straight without strain, and bends are needed to let the tube **expand/contract with temperature and absorb vibration.** Use the old tube (or a hand-bent **wire template**) as a pattern, and start bends a reasonable distance from fittings so the nuts and sleeves can slide back. [p. 9-15/9-16]

Inspection & repair limits:

- Aluminum scratches/nicks **not deeper than 10% of wall thickness** (and not in the heel of a bend) can be burnished out by hand. [p. 9-14]
- A **dent under 20% of tube diameter** is acceptable (unless it's in the heel of a bend). Remove dents by drawing a **bullet** (ball/slug) through the tube. [p. 9-14]

- **Any crack or deformity in a flare = reject.** Replace lines with severe die marks, seams, or splits. [p. 9-14]
- Hard, thin-wall CRES and titanium have **much smaller** damage limits — consult the aircraft maintenance manual. [p. 9-14]

Support spacing: all fluid lines must be supported at set intervals (Figure 9-43 gives max distance between supports by OD & material — e.g., 1/4" aluminum every 13.5", same steel every 16"). [p. 9-23] Use **bonded clamps** for metal hydraulic/fuel/oil lines (remove paint/anodizing under the bonding clamp); unbonded clamps are for wiring only. [p. 9-23]

11. FLEXIBLE Hose — Materials & Construction

Pure rubber is never used. Synthetics give the needed strength, durability, and workability. The common ones: **Buna-N, neoprene, butyl, EPDM, and Teflon™.** Teflon is in its own category; the rest are synthetic rubbers. [p. 9-16]

Material	Good with	Do NOT use with
Buna-N	Excellent with petroleum products	Phosphate-ester fluid (Skydrol™). Don't confuse with Buna-S.
Neoprene (acetylene base)	Petroleum (not as good as Buna-N) but better abrasion resistance	Skydrol™
Butyl (from petroleum raw materials)	Excellent with Skydrol™ (phosphate-ester)	Petroleum products
Teflon™ (tetrafluoroethylene resin)	Nearly everything — fuel, petroleum/synthetic oils, alcohol, coolants, solvents. Temp range –65°F to +450°F ; almost unlimited shelf/service life	(broadest compatibility of all)

5th-grade memory hook: Buna-N and Neoprene love oil/gas but *hate* Skydrol. Butyl is the opposite — it loves Skydrol but *hates* oil/gas. Teflon is the easy-going one that gets along with almost everything.

Construction of rubber hose: a **seamless synthetic-rubber inner tube**, wrapped in layers of **cotton braid and wire braid**, with an outer layer of **rubber-impregnated cotton braid**.

[p. 9-16] **Teflon hose** is extruded into a tube then covered with **braided stainless-steel wire** for strength. [p. 9-17] Brake hoses sometimes get an extra stainless wire braid over the outside for protection. [p. 9-16]

11a. Rubber hose pressure classes

Class	Pressure	Reinforcement
Low pressure	below 250 psi	Fabric braid
Medium pressure	up to 3,000 psi (smaller sizes to 3,000; larger sizes to 1,500)	One wire braid
High pressure	all sizes up to 3,000 psi	Multiple wire braids

Pressure classes from p. 9-16. [p. 9-16]

12. Hose IDENTIFICATION & SIZE (the ID rule)

Printed on the hose is a **lay line** (a stripe running straight down the length) plus the **type, the quarter and year of manufacture, and a 5-digit manufacturer code**. The lay line marks the hose's **natural lay (no twist)** and is repeated at intervals of **no more than 9 inches**. [p. 9-16/9-17]
Hose suitable for phosphate-ester fluid is marked "**Skydrol use.**" [p. 9-17]

THE HOSE SIZE RULE (memorize the contrast): Hose is sized by a **dash number** that matches the tubing size it works with. The dash number is keyed to **INSIDE DIAMETER (ID)** — it does **NOT** denote OD. When the hose dash number equals the tube dash number, you have the right size. [p. 9-19]

Compare: **RIGID TUBE = OD in 16ths** [p. 9-2] vs. **HOSE = ID, by dash number** [p. 9-19]

13. Inspecting, Fabricating & Replacing Flexible Hose

13a. Inspection — signs of deterioration

At every inspection, look for: **leakage, separation of the cover/braid from the inner tube, cracks, hardening, lack of flexibility, or excessive "cold flow."** Any of these = replace the hose. [p. 9-18]

Define “cold flow”: the **deep, permanent impressions** left in a hose by the pressure of hose clamps or supports. Because of cold flow (a setting process), **recheck/retighten clamps for several days after install.** [p. 9-18 / 9-23]

- Hose with **swaged (factory) end fittings** that fails → **replace the entire assembly** (order new, correct size/length, factory fittings). [p. 9-18]
- Hose with **reusable end fittings** → you can **fabricate a replacement** using the manufacturer’s tooling/instructions. [p. 9-18]

13b. Fabricating a hose assembly (MS detachable fittings)

MS-type end fittings are **detachable and reusable** if serviceable. The fitting’s ID equals the hose ID. [p. 9-18] Steps (Figure 9-35):

1. Vise the hose; cut to length with a fine-tooth hacksaw or cutoff wheel. [p. 9-19]
2. (High-pressure hose only) slit the cover to the wire braid with a knife — don’t nick the braid — and twist the cover off with pliers. [p. 9-19]
3. Screw the **socket onto the hose counterclockwise.** [p. 9-19]
4. Liberally lubricate inside the hose and the nipple threads. [p. 9-19]
5. Screw the nipple into the socket; leave **0.005" to 0.031" clearance** between the nipple hex and the socket. [p. 9-19]

CAUTION on lubricant: for hose made for **synthetic fluids (Skydrol™ / HYJET)**, **do NOT use any petroleum product** as the assembly lubricant — use a **vegetable-soap liquid** instead. [p. 9-19]

Proof-testing: every hose must be **proof-tested after assembly** with internal pressure. Liquid (hydraulic oil/water) for hydraulic/fuel/oil; dry oil-free air or nitrogen for air/instrument lines (gas tests are done **underwater**). A repaired/overhauled hose should be tested to **at least 1.5× system pressure** before install. [p. 9-19]

14. Installing Flexible Hose Assemblies (projects G22/G23)

This is the hands-on part for the hangar. Four things to get right: SLACK, DO-NOT-TWIST, BEND RADIUS, and SUPPORT/CLAMPS.

- **SLACK:** never install hose with a mechanical load on it. Leave **5% to 8% slack/bend of total length** — pressurized hose **contracts in length and expands in diameter.** [p. 9-21]

- **DO NOT TWIST:** install without twisting (twisting can rupture the hose or loosen the nuts). **Use the lay line / linear stripe** to confirm no twist — a straight stripe means no twist; if it spirals, the hose is twisted. Swivel connections relieve twist stress. [p. 9-21 / 9-17]
- **BEND RADIUS:** never bend below the hose's **recommended minimum bend radius** — too-sharp bends drop the burst pressure far below rated. Use elbow fittings to relieve sharp bends. [p. 9-21]
- **FLEX / straight ends:** the hose must stay **straight for at least two hose diameters from the end fittings** (the metal ends don't flex). [p. 9-21]
- **SUPPORT:** support hose **at least every 24 inches** (closer is better); never stretch hose tightly between two fittings; keep clearance from all other lines/structure. [p. 9-21]
- **Preformed hose: never straighten a preformed hose assembly;** use a support wire if it must be removed. [p. 9-17/9-18]

14a. Hose CLAMPS & support clamps

Use a **hose-clamp torque-limiting wrench** when available (made in **15 and 25 in-lb** limits). With no torque wrench, use the **finger-tight-plus-turns** method (values are approximate — use judgment). [p. 9-22]

Situation	Tighten to
Initial install — self-sealing hose	~ 15 in-lb
Initial install — all other aircraft hose	~ 25 in-lb
Retighten in service — non-self-sealing	If clamp won't finger-tighten, leave it alone unless leaking; if leaking, tighten 1/4 turn
Retighten in service — self-sealing	If looser than finger-tight, bring to finger-tight and add 1/4 turn

Support clamp types: rubber-cushioned clamps secure lines subject to vibration (cushion stops chafing); **plain** clamps for non-vibration areas; **Teflon-cushioned** clamps where Skydrol/hydraulic fluid/fuel would attack rubber (but Teflon damps vibration less). [p. 9-22] Clamps smaller than the hose OD can **restrict flow** — use the correct size. [p. 9-23]

MUST KNOW — the verbatim test items for the Block exam

1. **37° flare (35°–37°) for aircraft tubing. Automotive 45° flaring tools are forbidden.** [p. 9-4]

2. **RIGID TUBE** is sized by **OUTSIDE DIAMETER (OD)**, in 16ths of an inch (No. 6 = 3/8"). **HOSE** is sized by **INSIDE DIAMETER (ID)**, by dash number.
[p. 9-2 / 9-19]
3. **DOUBLE FLARE** is used on soft aluminum-alloy tubing 3/8" OD and under.
[p. 9-4]
4. **Hazard markings:** **FLAM** = flammable (fuel); **TOXIC** = toxic; **PHDAN** = physically dangerous (oxygen/nitrogen/Freon). [p. 9-7]
5. **NEVER** use titanium in any oxygen system (oxygen-reactive, can ignite).
[p. 9-1]
6. **Bend flattening:** small diameter of a flattened bend must not be less than 75% of original OD. [p. 9-3]
7. **Flareless tube install:** final tighten 1/6 turn from bottoming; never exceed 1/3 turn total (two flats). [p. 9-12/9-14]
8. **Flareless presetting:** aluminum $\leq 1/2"$ OD = 1 to 1-1/6 turns; steel & aluminum $> 1/2"$ OD = 1-1/6 to 1-1/2 turns. [p. 9-6]
9. **Hose slack** 5–8%; support every 24 in; lay-line stripe = do not twist; straight for 2 hose diameters from end fittings. [p. 9-21]
10. **Aluminum color codes:** 1100 White, 3003 Green, 2024 Red, 5052 Purple, 6061 Blue+Yellow, 7075 Brown+Yellow. [p. 9-2]

15. Quick Glossary (define-on-first-use, all in one place)

- **Rigid line / tubing** — stiff metal pipe for still, straight runs. [p. 9-1]
- **Flexible hose** — bendy line for moving/vibrating spots. [p. 9-1]
- **OD / ID** — outside / inside diameter. $ID = OD - 2 \times \text{wall}$. [p. 9-2]
- **Flare** — funnel-shaped flared tube end that seals against a fitting. [p. 9-3]
- **Bead** — raised ring rolled near a tube end so a hose+clamp grips it. [p. 9-7]
- **Sleeve / ferrule** — the ring in a fitting that bites or supports the tube. [p. 9-9]
- **Presetting** — pre-biting the sleeve onto a flareless tube before install. [p. 9-5]
- **Anneal** — heat metal to dull red & quench to soften it (before double-flaring). [p. 9-4]
- **Lay line** — the straight printed stripe that proves the hose isn't twisted. [p. 9-16/9-21]
- **Cold flow** — permanent dents in hose from clamp/support pressure. [p. 9-18]
- **CRES** — corrosion-resistant (stainless) steel. [p. 9-1]
- **FOD** — foreign object damage. [p. 9-1]

This is a plain-English summary of FAA-H-8083-30 Chapter 9 (pp. 9-1 to 9-23) for make-up study. Numbers, turn-counts, and limits are quoted from the handbook with page cites. For any graded AIM worksheet, match the answer to the AIM slide wording if it differs; this guide uses FAA handbook wording, which is what the written exam grades against.

Block 3 — Chapter 1: Safety, Ground Operations & Servicing

Plain-English Study Guide

amt-tutor make-up study session · 2026-06-14 · taught at 5th-grade-teacher level, nothing dumbed-out

Source (read in full, source of truth): FAA-H-8083-30B (Aviation Maintenance Technician Handbook—General), **Chapter 1 — Safety, Ground Operations, and Servicing**, printed pages 1-1 through 1-30 (your file “Chapter 1 Ground Ops.pdf”). Every fact below is cited to its printed page (e.g. “p. 1-14”).

How to use this: read this instead of the whole chapter. The yellow MUST KNOW boxes are the items most likely to show up on your class quiz and the cumulative block exam — memorize those cold.

PART 1 — Shop Safety

The whole chapter starts with one idea: a clean, orderly shop is a safe shop. Keep the shop, hangars, and flight line clean; the outgoing shift stows tools, stands, hoses, and cords. Signs mark dangerous equipment and where first-aid/fire gear is. Painted **safety lanes, walkways, and fire lanes** keep foot traffic out of work areas. [p. 1-2]

Electrical Safety

Electricity through the human body causes **burns at the point it enters and the point it exits**, and it can damage or destroy the **nervous system**. [p. 1-2] To work safely you need a working knowledge of electricity AND a healthy respect for it.

- **Two attitudes cause electrical accidents: fear and overconfidence.** A little respect is healthy; too much of either is deadly. Lack of respect usually comes from lack of knowledge. [p. 1-2]
- Safety gear that protects you (and gives peace of mind): **rubber gloves, safety glasses, rubber or grounded safety mats.** [p. 1-2]
- **Electrical fire prevention:** keep the area clean, uncluttered, and free of flammables. Keep cords free of kinks/bends. Never put cords where they get walked on or run over. If wires inside a cord break, the remaining wires carry more current → more heat → fire. [p. 1-2]

5th-grade picture: A wire is like a water hose. If half the strands inside break, the leftover strands have to carry all the “water” (current) — they get overworked and hot, just like pinching a hose makes pressure spike. Too hot = the insulation melts and starts a fire.

Safety Around Compressed Gases

Compressed air, like electricity, is a great tool only when it is under control. [p. 1-2] The dos and don'ts:

- Inspect air hoses often for breaks/worn spots; replace bad hoses immediately. Keep connections in a “no-leak” condition. [p. 1-2]
- **Never use compressed air to clean hands or clothing** — pressure forces debris into the flesh and causes infection. Never spray it toward other people. [p. 1-2]
- Coil and store hoses when not in use. Filter air used for paint spraying to remove oil and water. [p. 1-2]
- **Tire mounting / inflation:** many compressed-gas accidents happen mounting tires. Use tire dollies for heavy tires. When inflating, **always use tire cage guards** and a **pressure regulator** on high-pressure bottles to avoid over-inflation. **Tire cages are NOT required when adjusting pressure in tires still installed on an aircraft.** [p. 1-3]

Safety Around Hazardous Materials — where the hazard info lives

Two things tell you a chemical's danger:

1. **The risk diamond (material safety diamond)** — a four-color segmented diamond: **red = flammability, yellow = reactivity, blue = health, white = special hazard.** The red/yellow/blue blocks hold a number **0 to 4** (0 = little/no hazard, **4 = very hazardous**). The white block holds a word/abbreviation: RAD = radiation, ALK = alkali, Acid = acidic, CARC = carcinogenic, and a **W with a line through it = reacts strongly with water.** [p. 1-3]
2. **The SDS (Safety Data Sheet)** — the detailed version: full chemical breakdown, formulas, and what to do if you contact the chemical. **Formerly called the MSDS** (Material Safety Data Sheet). The required information is the same on every sheet, but its exact location can vary by manufacturer. SDSs are required by the governing safety body, **OSHA** (the U.S. Department of Labor's Occupational Safety and Health Administration). [p. 1-3]

MUST KNOW **Where to find hazard info:** quick look = the **risk diamond** (red flammability / yellow reactivity / blue health / white special, numbered **0–4**, 4 = worst). Detailed info = the **SDS** (Safety Data Sheet), **formerly the MSDS**, required by **OSHA**. [p. 1-3]

Safety Around Machine Tools

Each machine has its own rules, but they share a backbone: **wear eye protection, secure the work, set the correct RPM, and clean up after.** [pp. 1-3 to 1-4]

Machine	What it does	Key safety points
Drill press		

Machine	What it does	Key safety points
	Bore/ream holes, facing, milling	Eye protection; clamp work securely; set proper RPM; don't feed spindle past its travel limit; stop the machine before adjusting or clearing jammed work.
Lathe	Turns cylindrical work (inside or outside)	Eye protection; sharp tools; let the chuck stop on its own — never stop it by hand; don't set tools on the lathe; let work stop before measuring.
Milling machine	Shape/dress, cut gear teeth, slots, keyways	Eye protection; clean and secure the work bed; don't change feed speed while working; lower the table before moving away.
Grinder	Sharpen tools, dress metal, remove small amounts	Eye protection even with a shield; inspect the wheel for defects first; never force a wheel onto the spindle (it can explode); flanges/washer = 1/3 the wheel diameter; don't stand in the wheel's arc in case it explodes.

Welding (a special shop hazard)

Weld only in **designated areas**; remove the part from the aircraft if possible and weld in a controlled shop. If you must weld in the hangar: no open fuel tanks / fuel-system work, no painting, **no aircraft within 35 feet**, no flammable materials nearby, only qualified welders, rope off and placard the area, keep **fire extinguishing gear rated at least 20B in the immediate area with 80B as backup**, a trained fire watch present, and the aircraft kept towable with a tug attached and brakes released. [\[p. 1-4\]](#)

PART 2 — Flight-Line Safety

Hearing Protection — why and when

The flight line is loud from many sources at once — aircraft, **APUs, fuel trucks, baggage equipment** — and combined that noise can cause hearing loss. [\[pp. 1-4 to 1-5\]](#)

- **External** protection = earmuffs / headphones. **Internal** protection = plugs that fit into the auditory (ear) canal. Both lower the sound reaching the eardrum. [\[p. 1-5\]](#)
- **When it is essential:** working with **pneumatic drills, rivet guns, or other loud tools.** Even **short exposure** to these high-frequency sounds can cause hearing loss; continued exposure *will*. [\[p. 1-5\]](#)

Foreign Object Damage (FOD) — what it is and how to prevent it

FOD is any damage to aircraft, personnel, or equipment caused by any loose object — broken runway concrete, shop towels, safety wire, etc. [p. 1-5]

- **Prevention:** keep ramp/operation areas clean, run a **tool control program**, and provide convenient receptacles for used hardware, shop towels, and consumables. [p. 1-5]
- **The turbine-engine reason it matters:** a running gas turbine creates a **low-pressure area in front of the engine that sucks loose objects in**; the exhaust can throw objects far enough to damage anything they hit. Never leave tools or items near a turbine intake. [p. 1-5]

MUST KNOW **FOD = Foreign Object Damage** = damage to aircraft, people, or equipment from **any loose object**. Prevent it with a clean ramp, a **tool control program**, and receptacles for trash/hardware. [p. 1-5]

Safety Around Airplanes (the danger zones)

- **Propellers:** never assume a taxiing pilot can see you. **Stay within the pilot's view** on the ramp. [p. 1-5]
- **Turbine intakes (suck zone) and exhaust (blast zone) are very hazardous.** [p. 1-5]
- No smoking / open flames anywhere near an operating aircraft. Watch for aircraft fluids that harm skin. Keep ground equipment spaced from the aircraft and secured so it can't roll into it; stow loose items. [p. 1-5]

Safety Around Helicopters — rotor approach rules

Every helicopter is different, so learn the type. When approaching while the blades turn: [p. 1-5]

1. **Check that the rotor head and blades are level** — that gives maximum clearance as you approach.
2. **Approach in view of the pilot.**
3. **Never carry anything tall (vertical height)** that the blades could hit — that damages the blade and injures you.
4. **Never approach a single-rotor helicopter from the rear** — the spinning **tail rotor is invisible**.
5. **Never cross from one side to the other by going around the tail. Always go around the NOSE.**

MUST KNOW **Helicopter golden rules:** approach **in the pilot's view**, when the rotor is **level**; **NEVER approach from the rear** (invisible tail rotor); to change sides, **go around the NOSE, never the tail**. [p. 1-5]

Fire Safety on the flight line

Maintenance uses spark-making tools, heat, and flammable liquids/gases, so the fire potential is high. **The key to fire safety is knowing what causes a fire, how to prevent it, and how to put it out.** [p. 1-5] That leads straight into Fire Protection.

PART 3 — Fire Protection

What a fire needs (the fire triangle)

Three things are required for a fire. **Remove any one and the fire goes out:** [p. 1-5]

1. **Fuel** — combines with oxygen in the presence of heat, releasing more heat.
2. **Heat** — speeds the combining of oxygen with fuel, releasing more heat.
3. **Oxygen** — the element that combines chemically with the fuel (oxidation). Rapid oxidation with noticeable heat and light = **combustion / burning**.

5th-grade picture: Think of a three-legged stool. Fuel, heat, and oxygen are the three legs. Knock out ANY one leg and the stool (the fire) falls over. That is literally how every extinguisher works — it removes one leg.

Note on the “tetrahedron”: some texts add a 4th item, the chemical chain reaction, to make a fire *tetrahedron* (4 sides). This FAA chapter teaches the **three-element triangle** (oxygen, heat, fuel) — that is what it pictures and what your worksheet follows. [p. 1-5, Fig. 1-3]

The classes of fire — A, B, C, D

The National Fire Protection Association (NFPA) names three basic classes (A, B, C); a fourth (D) covers burning metal. [p. 1-6]

Class	What burns	Best extinguisher
Class A	Ordinary combustibles — wood, cloth, paper, upholstery	Water (best); also CO ₂ and multipurpose dry chemical
Class B	Flammable petroleum products / liquids — gas, oil, grease, solvents, paints	CO₂, Halon (1211/1301), dry chemical, foam — NOT plain water (fuel floats on water)
Class C	Energized electrical wiring and equipment	CO₂ or Halon (non-conductive) — water only with extreme caution after power is OFF
Class D	Flammable metal — magnesium (shop, wheels/brakes), bad welding	Dry powder ONLY — NEVER water or CO ₂

MUST KNOW Fire classes (memorize letter → what burns): **A** = ordinary combustibles (wood/paper/cloth) · **B** = flammable liquids (fuel/oil/grease/paint) · **C** = energized electrical · **D** = flammable metal (magnesium). Class D is “not commercially a basic type” per NFPA because it’s caused by an A/B/C fire. [p. 1-6]

Extinguisher types — how each works and what to use it on

- **Water** — best on **Class A**. Two effects: deprives the fire of oxygen AND cools the burning material. **Not for Class B** (petroleum floats on water). Use on Class C only with *extreme* caution — all power off first, and beware residual charge in capacitors/coils. [p. 1-6]
- **Carbon dioxide (CO₂)** — for **Class A, B, and C**. Smothers the fire (depletes oxygen) and cools it. Self-expelling (it has enough pressure to push itself out). Cautions: parts get **extremely cold** — **frostbite risk**; in confined spaces it can suffocate the operator too. [pp. 1-6 to 1-7]
- **Halogenated hydrocarbon (Halon, industry “Freon”)** — most effective on **Class B and C** (can be used on A and D but less effective). Works by chemically interfering with combustion and blanketing out oxygen, and resists re-flash. [p. 1-7]
- **Dry powder** — effective on B and C but **best for Class D (metal) fires**. **Not recommended for general aircraft use** (except metal fires) because leftover residue/dust is hard to clean and damages delicate equipment. [pp. 1-7 to 1-8]

Exam-trap callouts (the wording the FAA loves):

- **NEVER use water OR CO₂ on a Class D (metal) fire**. Their cooling effect makes the hot metal expand explosively — metals burn at extremely high temperatures. [p. 1-6]
- **Carbon tetrachloride (Halon 104) is NOT approved for any fire-extinguishing use**. It is extremely toxic and, near hot metal, produces **phosgene gas** (plus hydrochloric acid vapor and chlorine). [p. 1-8]
- **Halon 1301** = Bromotrifluoromethane (vapor spray, harder to direct, least toxic, UL rating 6). **Halon 1211** = Bromochlorodifluoromethane (forms a spray like CO₂, UL rating 5). The EPA restricted Halon to its 1986 production level (ozone). [pp. 1-7 to 1-8]
- **Toxicity table**: a HIGHER UL/group number = LESS toxic (6 = least toxic). Don’t read it backwards. [p. 1-7, Fig. 1-6]

Inspection of fire extinguishers

Check extinguishers periodically with a checklist. If no checklist, at minimum verify: [p. 1-8]

- Proper location of the appropriate extinguisher · safety seals unbroken · all external dirt/rust removed

- Gauge/indicator in the operable range · proper weight · no nozzle obstruction · no obvious damage

Identifying an extinguisher (the markings)

Extinguishers are marked to show which fire class they suit, and the same marking goes in a conspicuous spot near the extinguisher. A location marked for Class B must actually hold a Class-B-suitable extinguisher. Markings (decals, paint) must be legible at the required distance — on the extinguisher itself, readable at **at least 3 feet**; on a wall panel, readable at **at least 25 feet**.

[pp. 1-8 to 1-9]

How to USE an extinguisher — the PASS method

The chapter spells out the steps; the standard memory word is **PASS**:

Letter	Step	The chapter's wording
P	Pull the pin	“Most extinguishers have a pin to pull that allows the handle to activate the agent.”
A	Aim at the base of the fire	“Stand back 8 feet and aim at the base of the fire or flames.”
S	Squeeze the lever	“Squeeze the lever...”
S	Sweep side to side	“...and sweep side to side until the fire is extinguished.”

MUST KNOW **PASS** = **P**ull the pin · **A**im at the **base** of the fire (from about **8 feet** back) · **S**queeze the lever · **S**weep side to side. And before all that: confirm it's the **correct type for that class of fire**. [p. 1-9]

PART 4 — Tie-Down Procedures

Preparing the aircraft: tie aircraft down after each flight to prevent storm damage. Park/tie-down direction is set by the prevailing or forecast wind — **head the aircraft into the wind**, allowing ample wingtip clearance between tie-downs. After it's positioned, **lock the nosewheel or tailwheel in the fore-and-aft position**. [pp. 1-9 to 1-10]

Aircraft type	How to secure it
Land plane (light)	Tie ropes only to the provided tie-down rings — never to a lift strut (a slipping rope can bend the strut). Manila rope shrinks when wet, so leave about 1 inch of slack — too much slack lets it jerk; too tight puts inverted-flight stress on the airframe. Use an anti-slip knot like a bowline . [p. 1-10]

Aircraft type	How to secure it
Land plane (heavy)	Rope or cable; number of tie-downs depends on weather. Most have surface control locks — install them; add control-surface battens for high wind. Procedure: head into wind → install control locks/covers/guards → chock all wheels fore and aft → attach tie-down reels (or 1/4" wire cable / 1-1/2" manila line). Use tie-down stakes for temporary only. [p. 1-11]
Seaplane	Moor to a buoy or tie to a dock (weather permitting); waves make it bob/roll, which can cause damage. In a storm you can't fly out of, flood the floats with water and tie down; if possible, remove from the water and tie like a land plane, or anchor in a sheltered area. [pp. 1-10 to 1-11]
Ski plane	Tie down like a land plane if anchors exist. On ice/snow, use a "dead-man" — any item (pipe, log) tied to a rope and buried in a packed snow/ice trench (pour water and let it freeze). You can also freeze the skis to the ice. Caution: make sure a ski isn't still frozen down before moving the aircraft. [p. 1-11]
Helicopter	Hangar it if possible; tied down it survives winds up to ~65 mph . Evacuate for tornadoes/hurricanes. Mooring: (1) face it into the highest forecast wind/gusts; (2) spot it slightly more than one rotor span from other aircraft; (3) chock all wheels (or lower onto skids and pin); (4) align blades and install tie-down assemblies per the manufacturer (snug, with slack in wet weather); (5) fasten ropes/cables to the forward and aft landing-gear cross tubes. Tie down the main rotor blades if high winds are expected. [pp. 1-11 to 1-12]
Weight-shift-control	Very wind-susceptible. In light wind, secure the wing like a conventional aircraft. In high wind, disconnect the mast from the wing, lay the wing near the ground and secure it , or partially disassemble / hangar it. [p. 1-12]
Powered parachute	Pack the parachute (chute) in a bag so wind can't fill it and drag the seat and engine. Secure the engine and seat too if needed. [p. 1-12]

MUST KNOW **Tie-down numbers:** head **into the wind**; manila rope needs about **1 inch of slack** (it shrinks wet); **never tie to a lift strut**; use a **bowline** (anti-slip) knot; helicopters survive **~65 mph** tied down and spot **> one rotor span** from other aircraft; seaplane storm move = **flood the floats**; ski plane = **"dead-man"** buried in the snow. [\[pp. 1-10 to 1-12\]](#)

PART 5 — Ground Movement of Aircraft

Before starting ANY aircraft engine (applies to all powerplants): [\[p. 1-13\]](#)

1. **Head the aircraft into the prevailing wind** for cooling airflow over the engine.

2. Make sure prop blast / jet exhaust won't cause property damage or injury.
3. If using external electrical power, make sure it can be removed safely and is enough for the whole start.
4. **Station a "fireguard" with a suitable fire extinguisher** — a CO₂ extinguisher of **at least 5-pound capacity**, placed adjacent to the outboard side of the engine, in the pilot's view.
5. For turbine aircraft, keep the area in front of the jet inlet clear of personnel, property, and debris (FOD).
6. Follow the manufacturer's checklists for start and shutdown.

Reciprocating engines — start cautions

- **Hydraulic lock:** before starting a radial engine that sat **more than 30 minutes**, ignition OFF, then **turn the prop 3–4 complete revolutions by hand** to detect a hydraulic lock (liquid pooled in a lower cylinder). A lock shows up as abnormal effort or the prop stopping abruptly. **Never force the prop** — you can bend or break a connecting rod. To clear it, **remove a lower-cylinder spark plug and pull the prop through**; never turn it backward (that pushes liquid into the intake). [p. 1-13]
- **Mixture:** general rule — **idle cut-off for fuel injection, full rich for float-type carburetors**. [p. 1-13]
- Throttle to a position giving **1,000–1,200 rpm**; leave carb heat / alternate air in **COLD** to prevent fire damage if it backfires. [p. 1-13]
- **Starter duty cycle:** don't crank continuously for more than **1 minute**; allow a **3- to 5-minute cooling period** between attempts or you can burn out the starter. [p. 1-14]
- After it starts: move mixture to full rich (if started in idle cut-off) and **check oil pressure — if it doesn't show within 30 seconds, shut down**. Warm-up rpm is usually **1,000–1,300 rpm**. [p. 1-14]
- Keep the **cowl flaps OPEN** during ground running (these engines are air-cooled and depend on airflow). [p. 1-14]

Hand-cranking (hand-propping) dangers

If there's no self-starter, you start by turning the prop by hand. The call-and-response is mandatory: [p. 1-14]

1. The propeller-turner calls: **"Fuel on, switch off, throttle closed, brakes on."** The operator checks each item and **repeats the phrase**.
2. Switch/throttle are NOT touched again until the prop-swinger calls **"contact."** The operator repeats "contact," THEN turns the switch on. **Never turn the switch on and then call "contact."**

Hand-propping traps to remember:

- **Always assume the ignition is ON** while touching a prop — a faulty switch can read OFF and still let current flow (magnetos work by short-circuiting the current to stop), so the engine could fire. [p. 1-14]
- Be sure the ground is firm (no grass/mud/grease/gravel). **Swing the blade downward with the palms**; never curl fingers over the edge — a **kickback** can break them or pull you into the blades. Stand close enough to step away as the blade comes down. [p. 1-14]
- **Over-priming** causes fires, scored cylinders/pistons, and hydraulic lock. If flooded: ignition OFF, throttle FULL OPEN, turn it over to clear fuel — if excessive force is needed, STOP. [p. 1-14]

Extinguishing engine fires (induction / exhaust fire response)

A fireguard with a **CO₂** extinguisher stands by during every start, and must know the engine's induction system so he/she can direct CO₂ into the **air intake**. [p. 1-15]

1. **If an induction fire starts during cranking, keep cranking to start the engine and blow out the fire.**
2. If it won't start and the fire continues, discontinue the start; the fireguard then extinguishes it with the available equipment.

Turboprop starting (the 3 steps every turbine start follows)

Every turbine start is **three steps in sequence**: (1) the **starter** spins the compressor to make airflow; (2) at the right speed the **igniters** fire a hot spark; (3) **fuel** is introduced and lights off. As it accelerates to a **self-sustaining speed**, the starter is disengaged. [p. 1-15]

- **Fixed turbine**: prop is connected directly, so it turns during start (extra drag). **Free turbine**: no mechanical link between gas generator and power turbine, so the prop stays feathered and only turns as the gas generator accelerates. [p. 1-15]
- **Starters**: small engines use an electric motor (or starter/generator); big engines use an **air turbine starter** fed by an APU, a ground air cart, or engine cross-bleed. [p. 1-15]
- **While starting, always**: observe the starter duty cycle; confirm enough air pressure / electrical capacity; **do NOT start if turbine inlet (residual) temperature is above the manufacturer's limit**; provide fuel under low pressure to the fuel pump. [p. 1-15]

Turbofan starting

Turbine aircraft need no preflight run-up unless investigating a malfunction. Remove covers, head into wind, clear the run-up area of personnel and loose items, check fuel sumps for water/ice, and inspect the inlet/fan blades for FOD and damage. [pp. 1-16 to 1-17] Typical sequence: power lever to idle → **boost pumps on** → confirm fuel inlet pressure → starter on (engine rotates, check oil

pressure) → ignition on → start lever to idle/start (fuel flows) → **light-off = a rise in exhaust gas temperature** → check N1 (fan) rotation and oil pressure → starter off at proper speed. [p. 1-17]

Unsatisfactory turbine starts (hot start / hung start)

Bad start	What happens	Cause & what to do
Hot start	Engine starts but exhaust gas temperature exceeds limits	Caused by an excessively rich fuel/air mixture — too much fuel or not enough airflow. Shut the fuel off immediately.
False / hung start	Engine starts normally but rpm stays at a low value and won't rise to normal starting rpm	Usually insufficient starter power , or the starter cut off before the engine self-accelerated. Shut the engine down.
Fails to start	Doesn't start within the time limit	Lack of fuel, no/insufficient power to the ignition exciter, or wrong fuel mixture. Shut down; in all unsatisfactory starts turn off fuel and ignition , then motor the compressor ~15 sec to clear fuel (or wait 30 sec to drain if you can't motor it).

MUST KNOW **Hot start** = EGT too high, from too-rich mixture → **shut fuel off NOW**.
Hung (false) start = engine lit but rpm **stuck low**, from weak starter → **shut it down**.
[p. 1-17]

Auxiliary Power Units (APUs)

An APU is generally a **small turbine engine** that supplies **compressed air for starting engines, cabin heating/cooling, and electrical power on the ground**. Turn the switch to start (spring-loaded), and it starts automatically while you watch exhaust gas temperature. APUs run at **idle at 100% rpm with no load**, and are normally used to start the main engines. [p. 1-17]

Towing of aircraft (wing walkers & limits)

Towing is hazardous if careless. Large aircraft are moved with a tow tractor ("tug") and a **tow bar**. [pp. 1-19 to 1-20]

- **A qualified person must be in the flight deck to work the brakes** in case the tow bar fails or unhooks. [p. 1-19]
- **Tow bars are built for tension (pulling) only** — not for torsional/twisting loads. [p. 1-19]
- **Wing walkers:** a wing walker is stationed at **each wingtip** to ensure clearance; a **tail walker** is added for sharp turns or backing. [p. 1-20]

- **Never walk or ride between the nosewheel and the towing vehicle**, and don't ride on the aircraft or tug or board/leave them while moving. [p. 1-20]
- **Towing speed must not exceed the walking team members' pace.** Don't start/stop suddenly; brakes are applied only on a team member's command in an emergency. [p. 1-20]
- Charge the brake system before towing; keep **chocks immediately available**; check tires and struts for proper inflation first. [p. 1-20]
- **Tail-wheel aircraft:** tow forward by attaching the bar to the **main landing gear**, and **unlock the tail wheel** first (or you can break the locking mechanism). **Tricycle gear:** tow forward by the **nosewheel axle**. [pp. 1-19 to 1-20]

Taxiing aircraft

Only rated pilots and qualified A&P technicians are authorized to start, run up, and taxi aircraft. [p. 1-20] A taxi signalman helps when the pilot's view is blocked; the signalman stands **slightly ahead of and in line with the aircraft's left wingtip** (the nose is on the signalman's left). Rule of thumb: **if you can see the pilot's eyes, the pilot can see your signals.** [p. 1-21] Day signals follow the FAA AIM; the night stop signal is the "emergency stop" (crossed lighted wands forming an "X"). [pp. 1-21 to 1-22]

Tower light	Meaning
Flashing green	Cleared to taxi
Steady red	Stop
Flashing red	Taxi clear of the runway in use
Flashing white	Return to starting point
Alternating red and green	Exercise extreme caution

Standard taxi light signals. [p. 1-21, Fig. 1-22]

PART 6 — Servicing Aircraft

Before servicing any aircraft fluid, consult the specific maintenance manual for the proper equipment and procedures. [p. 1-22]

Oil and fluids

- Engine oil is checked with a **dipstick or sight gauge**. **Reciprocating engines: check after the engine has been inactive. Turbine engines: check just after shutdown.** [p. 1-22]
- **Dry-sump systems hide oil** that seeped from the tank into the gear case — it won't show on the dipstick until the engine is run/motored. If you service before that oil returns, you'll

overfill. Never overfill the oil tank — oil foams as it circulates, and the tank's expansion space is left for that foaming. Always use the correct oil for the engine. [p. 1-22]

- Spilled oil/fuel/hydraulic fluid on skin or clothes must be removed ASAP (fire and health). [p. 1-22]
- **Hydraulic reservoir:** use the correct fluid (identify by container or color). Some reservoirs are air-pressurized — bleed the pressure before servicing. If you change filters, ensure system pressure is OFF first; after major fluid loss, **purge/bleed air** and check for leaks. [pp. 1-23 to 1-24]

Servicing with air / nitrogen

When servicing tires or struts with **high-pressure nitrogen**, use caution: **clean the area before connecting the filling hose and do not over-inflate.** [p. 1-24]

Ground Support Equipment (GSE)

- **Electric ground power units (GPUs)** come as **towed, stationary, or self-propelled.** Supply **constant-current/variable-voltage DC for turbine starts** and **constant-voltage DC for reciprocating starts.** Position the unit so it can't collide with the aircraft if its brakes fail; extend the cable near full length but don't stretch it. **Never move a power unit while service cables are attached or the generator is engaged.** [pp. 1-24 to 1-25]
- **Hydraulic ground power units ("hydraulic mule")** drain, filter, refill, and test the aircraft hydraulic system. **At 3,000 psi a leak stream can cut like a sharp knife** — inspect lines for cuts/frays and keep them kink-free. [p. 1-25]
- **Ground support air units (air carts)** provide **low-pressure (up to 50 psi), high-volume air** for engine starting and cabin heating/cooling. [p. 1-25]

Oxygen servicing equipment & OXYGEN HAZARDS

- **Two people are required to service gaseous oxygen:** one at the equipment control valves, one watching the aircraft system pressure, in communication. [p. 1-25]
- **Do NOT service oxygen during fueling, defueling, or any work that could be an ignition source.** Service oxygen **outside the hangar.** [p. 1-26]
- Oxygen comes as **gaseous** (steel cylinders) or **liquid (LOX)** (stored/converted in a liquid-oxygen converter). Only "**Aviator's Breathing Oxygen**" (**Federal Spec BB-o-925A, Grade A**) goes in aircraft breathing systems — NOT industrial (impurities) or medical (water that freezes). [p. 1-26]
- **LOX causes severe "burns" (frostbite) on skin contact.** [p. 1-26]

MUST KNOW OXYGEN — NO OIL/GREASE RULE: gaseous oxygen is chemically stable and nonflammable BY ITSELF, but it makes other materials **burn faster and hotter**, and it **combines with oil, grease, or bituminous material to form a highly EXPLOSIVE**

mixture that's sensitive to impact or compression. So **keep oil and grease completely away from oxygen** and only let qualified people service it. [p. 1-26]

5th-grade picture: Plain air is about 1/5 oxygen. Pure oxygen is like pouring rocket-fuel air on a flame — a tiny spark that would barely burn in normal air becomes a roaring fire. Add a smear of oil or grease and it can **explode just from being squeezed or bumped**. That's why "no oil near oxygen" is an iron rule.

PART 7 — Fuel Servicing

Types of fuel and identification

Two general types: **aviation gasoline (AVGAS)** for reciprocating engines, and **turbine fuel / jet fuel** for turbine engines. AVGAS grades are written as two numbers = the **lean-mixture and rich-mixture octane ratings** (e.g. 80/87 = 80 lean, 87 rich). Grades are also color-coded. [p. 1-26]

Fuel	Color	Used in / notes
AVGAS 80 (80/87)	 Red	Reciprocating engines (lowest lead)
AVGAS 100 (100/130)	 Green	Reciprocating engines
AVGAS 100LL (low lead)	 Blue	Reciprocating engines — the common general-use grade today
AVGAS 115 (115/145)	 Purple	Limited use in large reciprocating-engine aircraft
Jet A / Jet A-1	 Black band (fuel itself clear/straw)	Turbojet/turboshaft. Made from kerosene . (Jet B = kerosene + AVGAS blend.)

The fuel's color must match the color band on the piping and fueling equipment. [p. 1-26, Fig. 1-31]

MUST KNOW **AVGAS color code (memorize both directions): 80 = Red · 100 = Green · 100LL = Blue · 115 = Purple.** Jet fuel = **Black** band on equipment (actual fuel is clear/straw). The two AVGAS numbers = **lean / rich octane ratings**. [p. 1-26]

Exam trap — never mix fuels: Before ever mixing AVGAS and turbine fuel, check the Type Certificate Data Sheet. **Jet fuel in AVGAS** → less power and possible **detonation** (engine

damage, loss of life). **AVGAS in jet fuel** → **lead deposits** in the turbine and reduced service life. [p. 1-26]

Contamination control (water, solids, microbes)

Contamination is anything in the fuel that shouldn't be there. The three types: **water, solids, and microbial growth**. The best control is to **prevent it from getting in**; filters, separators, and screens catch most of the rest. [p. 1-26]

- **Water** — two forms: **dissolved (vapor)** and **free water**. Dissolved water isn't a big problem until cold turns it into free water that can form **ice crystals and clog filters**. Free water shows as **water slugs** (concentrations you drain after fueling) or **entrained water** (suspended droplets that make fuel look cloudy and settle out over time). [pp. 1-26 to 1-27]
- **Solids** — insoluble bits: rust, dirt, sand, gasket material, lint, shop-towel fragments. Particles as small as **1/20 the diameter of a human hair** can damage fuel controls. [p. 1-27]
- **Microbiological growth** — micro-organisms living in the **free water in jet fuel**; they grow in warm temperatures with iron oxide and mineral salts present. They form **slime/sludge that fouls filters**, emulsify the fuel, and make **corrosive compounds** that attack the tank. **The best prevention is to keep the fuel dry.** [p. 1-27]

Fueling hazards — volatility & static

- **Volatility** = a liquid's ability to change into a gas at a relatively low temperature. **Liquid fuel doesn't burn — the VAPOR does.** That vapor is what powers the engine and what creates the fire hazard. [p. 1-27]
- **Static electricity** is made by one substance rubbing another — fuel flowing through a line, and especially an aircraft moving through the air in flight, builds a static charge. If that charge isn't dissipated before refueling, it tries to return to ground through the fuel line, and **the spark can ignite fuel vapor.** That's why we bond and ground. [p. 1-27]

Static / bonding & grounding during fueling (the order matters)

When fueling with mobile (truck) equipment, the chapter spells out the sequence: [pp. 1-27 to 1-29]

1. Approach with caution; position the fuel truck so it can **drive away forward** (no backing) if needed. Set the hand brake and chock the wheels.
2. **Ground the aircraft, then ground the truck, then bond them together** with a connecting wire (three separate ground wires, or a "Y" cable).
3. Make sure grounds touch **bare metal or proper grounding points. Do NOT use the engine exhaust or propeller as grounding points.**
4. **Ground the nozzle to the aircraft, THEN open the fuel tank.**
5. Protect the wing and other items from spilled fuel and careless handling.
6. Check the fuel cap for proper installation/security before leaving.

7. **Remove the grounding wires in the REVERSE order** they were attached. (If the aircraft won't fly/move soon, the aircraft ground wire may be left attached.)

MUST KNOW **Bonding & grounding order:** ground the **aircraft** → **then the truck** → **bond them together** → **then ground the nozzle to the aircraft** → **then open the tank**. Remove the wires in **reverse**. **Never** ground to the exhaust or propeller. The reason for all of it: **a static spark can ignite fuel vapor**. [pp. 1-27 to 1-29]

Fueling procedures & the pre-fuel checklist

Two basic methods: small aircraft use the **over-the-wing** method (hose into ports on top of the wing); large aircraft use **single-point (pressure) fueling** — receptacles in the bottom leading edge fill all tanks at once, which is faster, limits contamination, and **reduces the chance of static igniting the fuel**. [p. 1-27]

Before fueling, the person fueling must: [pp. 1-27 to 1-29]

- Turn **all aircraft electrical systems and electronic devices (including radar) OFF**.
- Carry **nothing in shirt pockets** (it could fall in the tank) and **no flame-producing devices**.
- Verify the **proper type and grade of fuel** — **do not mix AVGAS and JET**.
- Ensure **all sumps have been drained**. Wear **eye protection** (gloves/apron help too).
- Don't fuel if nearby aircraft could blow dirt onto the open tank. **Don't fuel with lightning within 5 miles. Don't fuel within 500 feet of operating ground radar.**

MUST KNOW **Fueling distance limits:** no fueling with **lightning within 5 miles**, and no fueling **within 500 feet of operating ground radar**. Turn **all electrical/electronic gear (incl. radar) OFF** and drain the **sumps** first. [pp. 1-27 to 1-29]

Defueling

Procedures differ by aircraft — **always check the maintenance/service manual first**. Defuel either by **gravity** (need a way to collect the fuel) or by **pumping** (don't damage the tanks; the removed fuel **cannot be mixed back with good fuel**). General precautions: [p. 1-29]

- **Ground the aircraft and the defueling equipment.**
- **Turn off all electrical and electronic equipment.**
- Have the **correct type of fire extinguisher available**.
- **Wear eye protection.**

One-breath recap of the whole chapter: Keep the shop clean and respect electricity, compressed gas, and machine tools (eye protection always). On the flight line: hearing protection near loud tools, control **FOD**, and respect prop/intake/exhaust and rotor danger zones (helicopters: in the pilot's view, never from the rear, go around the nose). Fire needs **oxygen + heat + fuel**; classes **A/B/C/D** each get a matching extinguisher (water=A, CO₂=A/B/C, Halon=B/C, dry powder=D); use it with **PASS**. Tie aircraft **into the wind**. Start engines with a **CO₂ fireguard**; watch for hydraulic lock, hand-prop discipline, and turbine **hot/hung starts**. Tow with **wing walkers** and brakes manned. Service fluids per the manual; **no oil/grease near oxygen**; know **AVGAS colors** (80-red, 100-green, 100LL-blue, 115-purple, jet-black) and **bond/ground** before fueling.

Built by amt-tutor from the assigned chapter read in full (FAA-H-8083-30B Ch.1, pp. 1-1 to 1-30). This is a study summary, not a sign-off for real maintenance. Worksheet answers should match your AIM slide wording where it differs; the written-exam wording follows the FAA handbook as cited above on every fact.

Block 3 — Chapter 14: Human Factors

Plain-English Study Guide • FAA-H-8083-30 Chapter 14 • prepared for Matthew Johnson

Read this instead of the whole chapter. Every fact is cited to the printed page (e.g. “p. 14-9”) so you can jump back to the book. **Yellow MUST KNOW boxes** are the highest-yield exam items. **Purple boxes** are things the task asked for that are *not actually printed in this -30 chapter* (Aloha 243, IM SAFE) — worth knowing, but confirm with your instructor before trusting them on a chapter-graded worksheet. This is a summary only — no quiz attached.

1. What “Human Factors” Means — and Why It Matters

Human factors = the study of how people interact with their environment, and everything about us (our bodies, our minds, our limits) that affects how well we do a job (p. 14-1). Think of it like this: the airplane is only as safe as the *person* turning the wrench. Human factors is the science of keeping that person from making mistakes.

A **human error** is an **unintentional act of doing a task wrong** that can degrade the system (p. 14-1). The chapter lists three types of human error:

- **Omission** — NOT doing an act or task (you forgot a step).
- **Commission** — doing a task **INCORRECTLY** (you did it wrong).
- **Extraneous** — doing a task you were NOT authorized to do.

And four consequences of human error (p. 14-1): little/no effect → damage to equipment → personal injury → catastrophic.

MUST KNOW — the big number

About **80% of maintenance errors involve human factors** (p. 14-1), and overall **~80% of all aviation accidents are caused by human factors, ~20% by machine/technical failure** (p. 14-29, Fig 14-36). Of all accidents, roughly **75–80% are human error**, and about **12% of those are maintenance related** (p. 14-1). The exam loves these numbers. The whole point of the chapter: *the next big safety gains come from fixing the human, not the technology.*

Why maintenance errors are sneaky: a pilot’s mistake shows up fast and obvious. A mechanic’s mistake is often **latent** — hidden, sitting quietly, sometimes for a long time before it bites (a missed crack, a part installed wrong) — but it can be just as deadly (p. 14-1).

2. A Short History of Human Factors

You don’t need to memorize all of this — but know the milestones the chapter calls out (p. 14-6 to 14-9):

- **~1487 — Leonardo da Vinci.** His *Vitruvian Man* is one of the earliest guides to **anthropometry** (the study of body dimensions). He studied birds to design a flying machine (p. 14-6).
- **Early 1900s — Frank & Lillian Gilbreth** (industrial engineers). Created the **challenge-response / call-back** system (doctor says “scalpel,” nurse repeats “scalpel”). That same speak-it-out-loud protocol is used in aviation today. They also researched **fatigue** (p. 14-6).
- **1903 — Wright Brothers** pioneered human-factors thinking, designing controls a human could actually fly (p. 14-6).
- **WWI & WWII** — focus shifted from *fitting the man to the machine* to *designing the machine for the man* (human capabilities & limitations). The **Hawthorne Effect** showed motivation strongly affects performance (p. 14-7).
- **1978 — Airline Deregulation** brought “do more with less” pressure to maintenance (p. 14-8).
- **1988 — the FAA gets formally involved** with the first Human Factors Issues in Aviation Maintenance conference (p. 14-1). Resource: hf.faa.gov.
- **1990s — “culture change”** at U.S. carriers; 1991 Continental began CRM-type training in maintenance (p. 14-9).

NOT IN THIS CHAPTER — but classic HF history (instructor may mention)

The famous event usually credited with waking up **maintenance** human-factors awareness is **Aloha Airlines Flight 243 (April 28, 1988)** — a Boeing 737 whose fuselage skin tore open in flight from undetected fatigue cracking and corrosion (multiple-site damage). It is the textbook “why maintenance HF matters” case and lines up with the FAA’s 1988 start date above. **This specific event is NOT named in this -30 chapter**, so your AIM worksheet/exam graded against this chapter will not require it — but it’s worth knowing for the FAA written and for context. The chapter’s own real-world cases are Alaska 261, Lufthansa A320, Excalibur 320, China Airlines 611, Colgan Air, and American 1400 (*Fig 14-32, 14-33, 14-37*).

3. The PEAR Model — the “what to worry about” checklist

People • Environment • Actions • Resources

PEAR is a **memory jogger** (mnemonic) for the four big things a maintenance human-factors program should care about (*p. 14-9*). Easiest way to remember it: a **PEAR** is the fruit, and it reminds you of four buckets.

Letter	Stands for	Plain English — what it covers
P	People	The folks doing the work — their physical, physiological, psychological, and psychosocial sides: body size, age, strength, knowledge, training, attitude, fatigue, health, and how they get along with others. Includes planned rest breaks . (<i>p. 14-9, Fig 14-12</i>)
E	Environment	TWO environments: the physical one (heat, humidity, light, noise, the ramp/hangar/shop) AND the organizational one (the company’s culture, communication, supervision, morale, safety culture). (<i>p. 14-9, 14-10, Fig 14-13</i>)
A	Actions	The exact steps a job takes. The tool here is JTA = Job Task Analysis — it identifies the knowledge, skills, and attitudes each task needs. Many authorities require the JTA to be the basis of the company’s general maintenance manual and training plan. (<i>p. 14-10, Fig 14-14</i>)
R	Resources	Anything a tech needs to get the job done — tools, lifts, test equipment, manuals, work cards, parts, training, even a phone or protective clothing. The R element is about spotting the need for more/better resources . (<i>p. 14-10, Fig 14-15</i>)

MUST KNOW

PEAR = People, Environment, Actions, Resources. Environment has two halves (physical + organizational). The Actions tool is the **JTA (Job Task Analysis)**.

4. MEDA — the error-investigation tool

Maintenance Error Decision Aid

MEDA is a **Boeing-developed tool for investigating maintenance errors** (*p. 14-10*). Its whole idea: **find the factors that caused the error and fix them — don’t just blame and re-train the person**. Old-school investigations asked “WHO messed up?” so the real causes never got fixed and the mistake kept coming back. MEDA breaks that “blame and train” cycle.

MEDA rests on three principles (*p. 14-11*):

- **Positive employee intent** — techs want to do a good job and don’t make mistakes on purpose.
- **Contribution of multiple factors** — there’s usually a *chain* of factors, not one.
- **Manageability of errors** — most contributing factors **CAN** be managed.

The five MEDA steps (*p. 14-11*): **1) Event** (pick the error to investigate) → **2) Decision** (was it maintenance related? if yes, continue) → **3) Investigation** (use the MEDA results form) → **4) Prevention strategies** (implement & track fixes) → **5) Feedback** (tell the floor what changed so techs know).

5. The SHEL Model — the four things a person interacts with

Software • Hardware • Environment • Liveware

SHEL puts the **worker (Liveware)** in the **center** and looks at how that person interacts with four things. It was created by Professor **Elwyn Edwards (1972)** and modified by Capt. **Frank Hawkins (1975)** (p. 14-11, 14-12, Fig 14-16). It only looks at interactions that *involve a human*.

Letter	Stands for	Plain English — careful, “Software” is a trap
S	Software (Procedures)	NOT computer programs! Here “software” means the paperwork & procedures — manual layout, checklists, symbols, language, work cards. (p. 14-12)
H	Hardware (Machines)	The physical stuff — location of components, accessibility of parts and tooling. (p. 14-12)
E	Environment (Ambient)	Temperature, humidity, sound, light, and time-of-day. (p. 14-12)
L	Liveware (People)	The technician interacting with other people — managers, peers, family, friends, and self. This is the Liveware-to-Liveware interface : human-to-human. The worker in the center is also Liveware, so SHEL has a Liveware-Liveware link. (p. 14-12)

MUST KNOW — exam trap

In SHEL, **S = Software = procedures/paperwork, NOT computer code. L = Liveware = people.** The **Liveware-to-Liveware interface** is human-to-human interaction (the worker with managers, peers, family, and self). Worker sits in the *center*.

6. The Swiss Cheese Model (Reason’s Model of Accident Causation)

This is **James Reason’s Model of Accident Causation** (introduced 1990, revised 1993) — nicknamed the “**Swiss cheese**” model (p. 14-12).

The analogy: picture several slices of Swiss cheese standing in a row — each slice is a safety defense (a procedure, an inspection, an inspector). Every slice has **holes** (weaknesses). Most of the time a hole in one slice is blocked by solid cheese in the next slice, so nothing gets through. An accident only happens when the **holes in ALL the slices line up** and a hazard passes straight through every defense (p. 14-12).

Two kinds of failures make those holes (p. 14-12, 14-13):

Type	What it is	Example from the chapter
Active failure	An error whose effect is immediate and obvious — the sharp-end event.	An aircraft slips off a lifting jack because the tech placed it wrong (or: a tech falls off a ladder). (p. 14-12, 14-13)
Latent failure	A result of a decision/action made long before ; it can lie dormant a long time before the consequence shows.	Funding wasn’t approved to buy the correct jack, so a wrong jack was in use (or: the ladder was already broken and nobody replaced it). (p. 14-12, 14-13)

MUST KNOW

Swiss cheese = James Reason’s Model of Accident Causation. An accident happens only when the **holes in every slice (defense) line up**. **Active failure = immediate/obvious; latent failure = hidden, decided long before, lies dormant.** An accident is a **chain of events**; break the chain at any link and you stop the accident.

7. Types of Human Error — and Error vs. Violation

The chapter splits errors into **unintentional** vs. **intentional** (p. 14-13):

Term	Plain English
Unintentional error	An accidental slip or wrong judgment — you did NOT mean to. Example: reading a torque value and transposing 26 into 62, or grabbing the wrong work card. It's still a mistake, but not on purpose.
Intentional error = a VIOLATION	When someone knowingly chooses to do something wrong — purposely deviating from safe practices, procedures, standards, or regulations. The chapter says an intentional error “should really be considered a violation.”

MUST KNOW — the difference

An **ERROR** is an **unintentional mistake**. A **VIOLATION** is **intentional** — you knew the rule and broke it anyway. Same wrong outcome, but the difference is **intent**.

Don't confuse those *active/latent* failures from Section 6 with *unintentional/intentional* here — different pairs. Active vs. latent = **when the effect shows**. Unintentional vs. intentional = **whether you meant to**.

8. THE DIRTY DOZEN — the twelve killers (HIGHEST YIELD)

The **Dirty Dozen** = **twelve human factors that most often lead to maintenance errors**. They were identified by **Transport Canada** after a rash of maintenance-related accidents in the late 1980s / early 1990s, then adopted industry-wide (p. 14-13). If you memorize ONE list for the exam, make it this one.

MUST KNOW — the twelve, in order

1 Lack of Communication • 2 Complacency • 3 Lack of Knowledge • 4 Distraction • 5 Lack of Teamwork • 6 Fatigue • 7 Lack of Resources • 8 Pressure • 9 Lack of Assertiveness • 10 Stress • 11 Lack of Awareness • 12 Norms (p. 14-2, 14-13).

The error	What it means (plain words)	The safety net (how to prevent it)	Page
1. Lack of Communication	Information not passed on, or passed on wrong, between techs, shifts, pilots, parts suppliers, or management. Worst at shift change when a half-done job is handed off.	Use logbooks and worksheets to record exactly what is done; do a clear oral AND written turnover at shift change; never assume the work was completed.	p. 14-13
2. Complacency	Over-confidence from doing a task so many times you stop really looking. “I’ve checked this a hundred times, it’s always fine” — so you skip the look.	Always EXPECT to find something wrong; never sign off on anything you did not fully check; always double-check your work.	p. 14-14
3. Lack of Knowledge	Not knowing the system, the current procedure, or that this model differs from one you know. Leads to a faulty repair with catastrophic results.	Only fix what you are trained to fix; make sure the maintenance manual you use is up to date; if you don't know how, ASK someone who does.	p. 14-14, 14-18
4. Distraction	Anything that pulls your mind off the task — a phone call, an aircraft towed by, a family worry. You come back and think you're further along than you are. About 15% of maintenance errors come from distractions.	When you return, go back THREE steps in the procedure to confirm where you left off; use a detailed checklist; never leave tools/ parts lying around — secure them first.	p. 14-14, 14-19
5. Lack of Teamwork	The crew doesn't agree on roles or look out for each other, so things fall through the cracks (a gear swing or ops check needs the whole team working together).	Keep lines of communication open; discuss specific duties when a job needs more than one person; always look out for co-workers with safety in mind.	p. 14-16, 14-20

6. Fatigue	Tiredness from poor sleep, overwork, stress, or working against your circadian rhythm (your body clock). It slows reaction time, memory, and judgment — and you often can't tell you're fatigued. Most dangerous: working alone tired, 1–3 AM.	Be aware of the symptoms in yourself and others; forfeit (hand off) complex tasks when exhausted; eat healthy, exercise, keep a regular sleep pattern. Get 8–9 hrs sleep.	p. 14-16, 14-21
7. Lack of Resources	Missing or low-quality parts, tools, manuals, test equipment, or staff to do the job right. Tempts you to improvise — which can be very costly.	Keep enough parts on hand and order ahead; NEVER substitute an incompatible part to get the job done; keep all equipment properly maintained. (Term: AOG = Aircraft On Ground.)	p. 14-18, 14-23
8. Pressure	The push (from schedule, management, or yourself) to do it faster without mistakes. Self-induced pressure = pressure you put on yourself that wasn't really there.	Make sure the pressure isn't self-induced; speak up if you need more time rather than rushing; ask for extra help when time is tight.	p. 14-22, 14-25
9. Lack of Assertiveness	Not speaking up when something looks wrong. Being assertive = stating your concern positively and productively (NOT aggressively). Staying silent has caused fatal accidents.	Give clear feedback when you see a risk; never compromise your standards; let co-workers voice opinions and accept corrective criticism.	p. 14-22, 14-26
10. Stress	Your subconscious response to demands placed on you. Stressors are physical (heat, noise, bad light, tight spaces), psychological (family, money, conflicts), or physiological (illness, hunger, lack of sleep).	Take time off or a short break when stressed; ask a co-worker to monitor your work; healthy eating, exercise, and enough rest lower stress.	p. 14-24, 14-27
11. Lack of Awareness	Failing to see all the consequences of an action, or what's going on around you ("tunnel vision"), usually from doing the same task so often you stop thinking.	Check whether your work conflicts with an existing mod or repair; always ask co-workers to check your work; even if highly skilled, have someone verify it.	p. 14-26, 14-28
12. Norms	"Norms" is short for "normal" — the unwritten rules of "how we do it here." Unsafe norms (shortcuts, working from memory, skipping paperwork) become accepted as standard and cause accidents.	Make sure everyone follows the SAME standard; remember just because it seems normal doesn't make it correct; the easiest way is often not the approved way.	p. 14-26, 14-29

The chapter calls these safety nets "MITIGATING THE RISK" and shows three for each of the twelve (the green boxes in Figures 14-18 through 14-31).

9. Personal Minimums — the "Am I fit to work?" self-check

A **personal minimums checklist** is a quick honest self-test a tech runs **before** starting work: *am I in good enough shape — body and mind — to do safe work right now?* If the answer is no, you don't start (or you hand the task off). This is the practical, personal side of the Dirty Dozen factors **Fatigue** and **Stress**.

The chapter's own self-care guidance (p. 14-16, 14-17, 14-24, 14-27): watch for fatigue symptoms in yourself; get **8–9 hours of sleep**; avoid complex tasks at the bottom of your circadian rhythm (the **1–3 AM** low); manage stressors; hand off complex work if exhausted; eat well, exercise, limit alcohol/tobacco.

NOT IN THIS -30 CHAPTER — but the classic FAA self-check (worth knowing)

The well-known FAA self-assessment is the **IM SAFE** checklist (from the Pilot's Handbook FAA-H-8083-25 / AC 60-22). It is the standard “personal minimums” acronym. Many instructors teach it alongside maintenance HF even though **this -30 chapter does not print it**:

Letter	Ask yourself...
I	Illness — Am I sick?
M	Medication — Am I on anything that affects me?
S	Stress — Am I overloaded by life/work pressure?
A	Alcohol — Have I been drinking? (none on the job)
F	Fatigue — Am I rested enough?
E	Emotion / Eating — Am I emotionally OK and properly fed?

How a tech uses it: run IM SAFE in your head before clocking onto a job. Any “yes” to Illness/Medication/Stress/Alcohol/Fatigue (or “no” to being emotionally steady and fed) is a flag to slow down, get help, or step away — because a maintenance mistake is latent and may not show up until someone is flying. **Confirm with your AIM instructor whether IM SAFE is on your graded material for this chapter — it is FAA-standard but not in this -30 text.**

10. The 5 Things to Remember + Quick Acronym Card

1. **~80% of maintenance errors are human factors** — not lack of skill. Fixing the human is the next big safety win.
2. **The Dirty Dozen** are the 12 most common human-factor traps (Transport Canada). Know them cold, plus a safety net for each.
3. **PEAR** = People, Environment, Actions, Resources — the four buckets to manage.
4. **SHEL** = Software (procedures, NOT code), Hardware, Environment, Liveware (people) — worker in the center.
5. **Swiss cheese** = Reason's model: accidents happen when the holes in every defense line up. Active error = obvious now; latent = hidden from a past decision.

Acronym / Model	Spelled out	One-line job
PEAR	People • Environment • Actions • Resources	Checklist of the 4 things an HF program manages (14-9)
SHEL	Software • Hardware • Environment • Liveware	The 4 things the worker (center) interacts with (14-11)
MEDA	Maintenance Error Decision Aid	Boeing tool to investigate errors, not blame people (14-10)
JTA	Job Task Analysis	Lists knowledge/skills/attitudes a task needs (the “A” in PEAR) (14-10)
AOG	Aircraft On Ground	A grounding problem; rush to get parts to return to service (14-19)
Dirty Dozen	12 human factors (see Section 8)	The 12 most common causes of maintenance error (14-13)
Swiss Cheese	Reason's Model of Accident Causation	Holes in defenses must line up for an accident (14-12)
IM SAFE	Illness • Medication • Stress • Alcohol • Fatigue • Emotion	Personal-minimums self-check (FAA-H-8083-25, NOT in this -30 ch.)